

The gut contractile organoid: a novel model for studying the gut motility regulated by coordinating signals between interstitial cells of Cajal and smooth muscles

Reviewed Preprint

v2 • June 19, 2025

Revised by authors

Reviewed Preprint

v1 • July 9, 2024

Rei Yagasaki, Ryo Nakamura, Yuuki Shikaya, Ryosuke Tadokoro, Ruolin Hao, Zhe Wang, Mototsugu Eiraku, Masafumi Inaba, Yoshiko Takahashi 

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan • WPI Nano Life Science Institute (WPI-NanoLSI), Kanazawa University, Kanazawa, Japan • Ushimado Marine Institute, Okayama University, Okayama, Japan • Graduate School of Science, University of Hyogo, Hyogo, Japan • Department of Bioscience, Okayama University of Science, Okayama, Japan • Institute for Life and Medical Sciences, Kyoto University, Kyoto, Japan • Department of Polymer Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Japan

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This **valuable** study reports the development of a novel organoid system for studying the emergence of autorhythmic gut peristaltic contractions through the interaction between interstitial cells of Cajal and smooth muscle cells. The authors further utilized the system to provide **convincing** evidence for a previously unappreciated potential role for smooth muscle cells in regulating the firing rate of interstitial cells of Cajal. The work will be of interest to those studying development and physiology of the gut.

<https://doi.org/10.7554/eLife.97860.2.sa4>

Abstract

The gut undergoes peristaltic movements regulated by intricate cellular interactions. How these interactions emerge in the developing gut remains poorly explored due to a lack of model system. We here developed a novel contractile organoid that is derived from the muscle layer of chicken embryonic hindgut. The organoid contained smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs; pacemaker) with few enteric neurons, and underwent periodic contractions. The organoid was formed by self-organization with morphological arrangements of ICCs (internal) and SMCs (peripheral), allowing identification of these cells in live. GCaMP- Ca^{2+} imaging analyses revealed that Ca^{2+} transients between ICC-ICC, SMC-SMC or SMC-ICC were markedly coordinated. Pharmacological studies further suggested a role of gap junctions in ICC-to-SMC signaling, and also possible mechanical feedback from SMC's contraction to ICC's pace-making activities. In addition, two organoids with different rhythm became synchronized when mediated by SMCs, unveiling a novel

contribution of SMCs to ICC's pace-making. The gut contractile organoid developed in this study offers a useful model to understand how the rhythm coordination between/among ICCs and SMCs is regulated and maintained during gut development.

Introduction

In the gut, ingested material is conveyed properly along the gut axis by the gut movements called peristalsis, which is recognized as wave-like propagation of a local constriction. The physiology of gut peristalsis has extensively been studied in adults, where the peristalsis plays pivotal roles in effective transportation and digestion/absorption of inter-luminal contents. And many cases of gut-related pathology are associated with peristaltic dysfunctions. The gut peristaltic movements are achieved by intricate regulations of intercellular functions among multiple cell types. At the site of origin of peristaltic waves (OPW), a local constriction emerges along the circumferential axis, and this is soon followed by a progressive wave of the contraction along the gut axis. During these processes, the circumferential constriction demands multiple SMCs to achieve simultaneous contraction. However, due to the complex structure of the gut, how such synchronization/coordination in SMCs is regulated remains largely undetermined.

It is known in vertebrates that the embryonic gut undergoes peristaltic movements even without experience of food intake. The embryonic gut therefore serves as a powerful model to study the intrinsic mechanisms underlying the peristalsis contrasting with the adult gut where ingested content influences the gut motility increasing complexity in analyses. We have recently reported using chicken embryos that sites of OPW are randomly distributed along the gut axis at early stages, and they later become confined to specific sites, and that this confinement of OPWs enables rhythmic and patterned peristaltic movements (Shikaya et al., 2022). One of the long-standing and important questions is how the synchronized/coordinated contraction is achieved and maintained.

In a long history of gut motility studies, it has been known that enteric nervous system (ENS), smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs) play important roles in peristaltic movements (Barajas-Lopez et al., 1989; Camborova et al., 2003; Chevalier et al., 2020; Huizinga et al., 1995; Kito et al., 2005; Liu et al., 1998; Rumessen & Thuneberg, 1996; Sanders et al., 1991; Takaki, 2003; Takayama et al., 2002; Thomsen et al., 1998). It has widely been accepted that: 1) at early embryonic stages, gut movement/peristalsis does not require ENS activity, 2) ICCs serve as a pacemaker dictating their rhythm to SMCs, 3) the contraction is executed by SMC and not by ICCs, since thick fibers of myosin are found solely in SMCs. A series of elaborate electrophysiological studies showed that slow waves (a type of changes in membrane potential characteristic of gut movements) occur spontaneously in ICCs but not in SMCs, and that these slow waves lead to voltage-dependent Ca^{2+} influx evoking action potential, which is somehow transmitted to SMCs to execute gut contraction in register with ICC's pace-making rhythm (Baker et al., 2021; Torihashii et al., 2002). The knowledge that ICCs act as a pacemaker was supported by compelling evidence obtained by c-Kit-deficient mouse mutants (W/W^V), in which ICC differentiation was severely affected leading to a failure of gut peristalsis (Huizinga et al., 1995; Torihashii et al., 1999). However, it remains largely unexplored how the intrinsic/spontaneous rhythm in a single ICC becomes synchronized among multiple ICCs which constitute intricate networks in the gut. It has also been under debate to what extent the gap junction contributes to cell-cell communications between ICCs, SMCs or ICC-SMC. One reason is that it has been difficult to stably maintain SMCs and ICCs in cell culture condition, and also to distinguish ICCs from SMCs in the living gut (these two types of cells originate from the same progenitor of splanchnopleural mesoderm during development). Thus, a novel model system has been awaited to circumvent these obstacles. Recently, it was reported that differentiation states of mouse hindgut-derived cells were successfully maintained for a relatively long period of time in a

serum-free culture medium (Wang et al., 2018 [↗](#)). In that study, many types of gut-derived cells including not only ENS, ICC, SMCs but also glial cells and serosa were observed, and this large and complex cell mass was seen to undergo rhythmic contractions *in vitro*.

It has widely been appreciated that organoids can serve as a powerful model and tool to circumvent such obstacles of organ complexity. Relatively simple structures of organoids allow analyses at higher resolution than *in vivo*, and also permit analyses of cell behaviors in three-dimensional (3D) environment, which might be different from behaviors *in vitro* confined to two dimensions. In the current study, we have developed a novel organoid called “gut contractile organoid” by culturing chicken hindgut-derived cells in a serum-free medium. The gut contractile organoid undergoes periodic contractions, and it is essentially composed of ICCs and SMCs, the former residing centrally whereas the latter peripherally, allowing distinction between the two cell types in living organoids. These advantages enabled GCaMP-live imaging of Ca^{2+} dynamics, and revealed coordinated oscillations of Ca^{2+} transients between ICC-ICC, SMC-SMC and SMC-ICC. Pharmacological studies further suggested a role of gap junctions in an ICC-to-SMC signaling, and also a possible mechanical feedback from SMC’s contractions to ICC’s pace-making activities. In addition, by regarding an organoid as a single oscillator unit and by placing these oscillators separately in a hydrogel mold, we found that two oscillators with different rhythm become synchronized when mediated by SMCs, supporting the notion of SMC’s contribution to ICC’s pace-making. The gut contractile organoid developed in this study must be useful to unveil the intrinsic mechanisms underlying the rhythm coordination and its maintenance between/among ICCs and SMCs during peristaltic movements in the embryonic gut.

Results

Spheroids were formed from muscle layer-derived cells of the embryonic hindgut

To develop a culture condition that would facilitate analyses of gut motility, we dissected the muscle layer (also called *tunica muscularis*) from the hindgut of chicken embryos of embryonic day 15 (E15) by removing the serosa and intestinal epithelium (mucosa) (Fig. S1 [↗](#)). The isolated muscle layer was dissociated into single cells to prepare 5.0×10^5 cells per culture dish. We started analyses with a culture condition with FBS-free medium and Matrigel as substrate as previously described for cultures in mice of gastrointestinal cells including serosa (Wang et al., 2018 [↗](#)). We tested three kinds of FBS-free media: DMEM/Ham’s F-12, Ham’s F-12, and Neurobasal media (see Materials and Methods). When cultured in DMEM/Ham’s F-12 or Ham’s F-12, dissociated cells formed very small aggregates containing several cells at day 1, the morphology of which did not change significantly until day 5 (Fig. 1A [↗](#)). In clear contrast, when cultured in the Neurobasal medium, cells formed clusters that were interconnected by elongated cells with neighboring clusters as early as day 1. These clusters grew as larger aggregates by day 3, and became spherical by day 5. Such spheroids were not observed in the conditions with Ham’s F-12 or DMEM/Ham’s F-12. We also tested different substrates, Poly-Lysine or collagen for dish coating with the Neurobasal medium, but neither one yielded spheroid. To know how the spherical aggregates were formed under the condition of Neurobasal medium and Matrigel, we obtained time-lapse images at two-hour intervals. Originally sparse clusters that were loosely connected by elongated cells merged each other forming progressively larger clusters (Fig. 1B [↗](#), Movie 1). In the following experiments, we focused on the spheroids formed under the condition of Neurobasal medium and Matrigel coating.

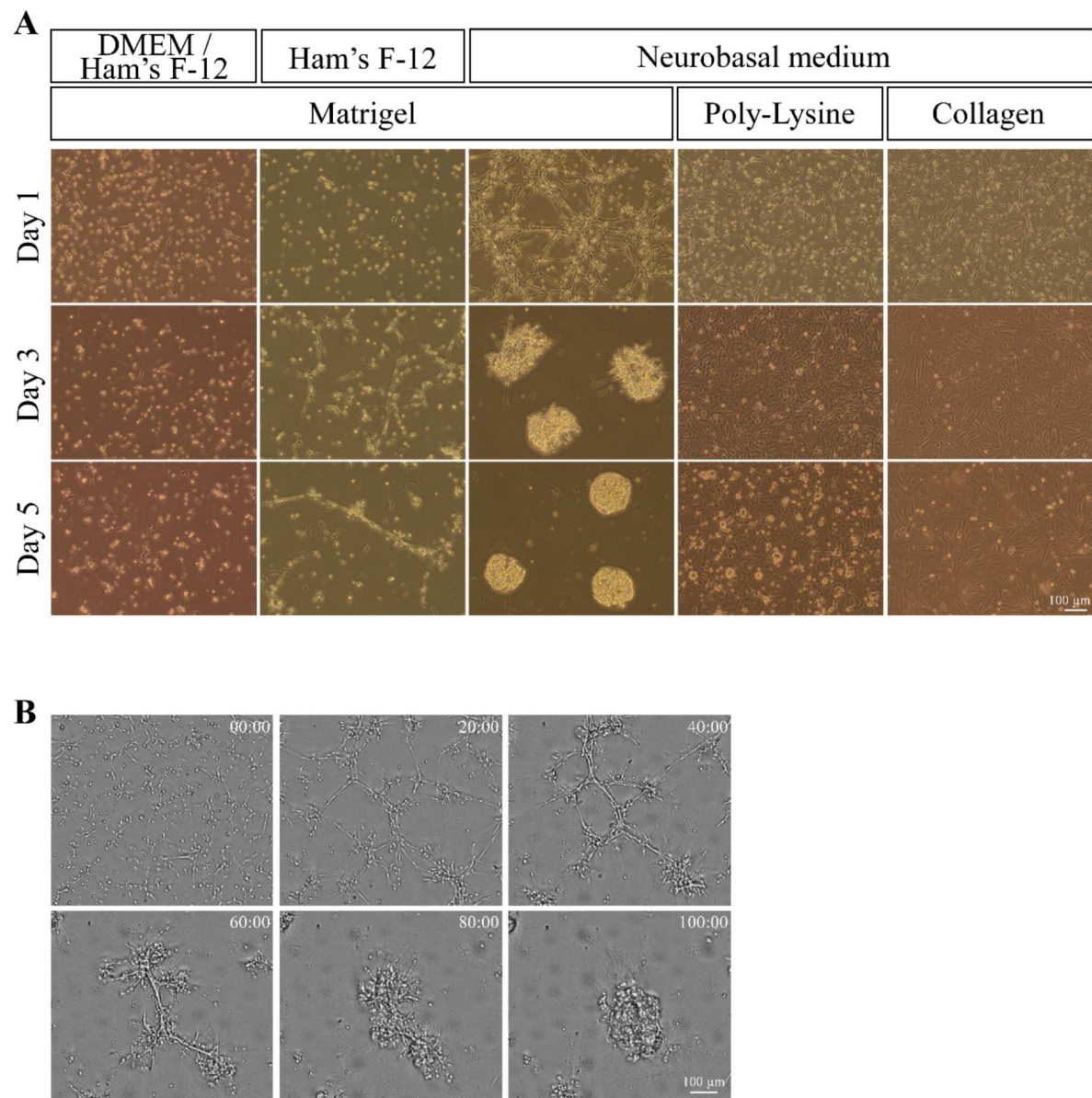


Figure 1.

Culture of muscle layer-derived cells prepared from embryonic hindgut.

(A) Culture of muscle layer-derived cells prepared from embryonic hindgut with FBS free-media and substrates. **(B)** Long-term time-lapse imaging after seeding on Matrigel with Neurobasal media. The images show the ability of these cells to self-assemble at 0, 20, 40, 60, 80 and 100 hours taken from Movie 1. Scale bars: 100 μ m (A, B).

The gut muscle layer-derived spheroids displayed periodic contractions

The spheroids underwent reiterative contractions at day 3, and these contractions were observed at least until day 7 of culture (**Fig. 2A** [for normalized intensities of 3 representative organoids for each stage, Movie 2](#)). Time-lapse imaging of the contractions combined with quantitative assessments by MATLAB (MathWorks, see Materials and Methods) showed that interval periods between two successive peaks were 13.3 seconds (sec), 15.4 sec and 19.6 sec for cultures at day 3, day 5 and day 7, respectively (median values) (**Fig. 2B** [; contraction intervals](#)).

The contracting spheroid was composed of ICCs and SMCs

To determine the cell types comprising the spheroids, we performed immunostaining with antibodies against the chicken c-Kit protein for ICCs ([Yagasaki et al., 2022](#)), and α SMA and desmin for SMCs. It is known that ICCs and SMCs are derived from common progenitors which are $c\text{-Kit}^+/\alpha\text{SMA}^+$, and that after differentiation ICCs and SMCs are $c\text{-Kit}^+/\alpha\text{SMA}^-$ and $c\text{-Kit}^-/\alpha\text{SMA}^+$, respectively ([Duband et al., 1993](#) ; [Kluppel et al., 1998](#)). In the clusters at day 3, $c\text{-Kit}^+/\alpha\text{SMA}^-$ signals were detected in the internal region, and $c\text{-Kit}^+/\alpha\text{SMA}^+$ signals at the periphery (**Fig. 3A** [, Day 3](#)), suggesting that internal cells were differentiated ICCs, whereas peripheral cells were ICC/SMC progenitors. At day 5 onward, $c\text{-Kit}^+/\alpha\text{SMA}^-$ cells and $c\text{-Kit}^-/\alpha\text{SMA}^+$ cells were segregated which were located internally and peripherally, respectively (**Fig. 3A** [, Day 5](#)), and this spatial segregation of the cells remained unchanged until day 7 (**Fig. 3A** [, Day 7](#)), the latest stage examined in this study. The peripherally lining cells were also positive for desmin staining, supporting that these cells were SMCs (**Fig. 3B**). Infection with RCAS-GapEGFP into a forming spheroid further visualized multipolar ICCs in the internal region (**Fig. 3C**), a morphology known to be characteristic of ICC-MY which would normally reside in the layer of myenteric plexus ([Mei et al., 2009](#) ; [Sanders et al., 2014](#)). ICCs of a different type were also recognized, which were elongated, thin, and lining underneath the peripheral SMCs (**Fig. 3A** [, Day 7](#)). It is possible that these were ICC-IM, known to be embedded in and tightly associated with smooth muscles in the gut ([Huizinga et al., 2011](#) ; [Iino & Horiguchi, 2006](#)). Unexpectedly, the spheroid contained few neural cells (ENS), if any, revealed by anti-Tuj1 antibody (**Fig. S2A, B**). Indeed, when tetrodotoxin was added to culture medium, contraction intervals of organoids were comparable to those of control (before the administration) (**Fig. S2C**) confirming little contribution by ENS. We counted cell numbers of each cell type per organoid, and found that contrasting with the intact gut, the proportion of SMCs was smaller than that of ICCs (**Fig. S2B**) possibly due to a cell type-specific loss during culture preparation.

Since ICCs and SMCs were spatially segregated in the sphenoid, we stained with anti N-cadherin antibody (E-cadherin is positive solely in the mucosa/endoderm) ([Graham et al., 2017](#) ; [Grosse et al., 2011](#)). N-cadherin signals were seen in the internal ICCs, and not in the peripheral SMCs (**Fig. 3D**). Thus, it is likely that N-cadherin plays a role in the segregation.

In summary, the hindgut-derived spheroid displays three prominent characteristics: (1) a dominant occupation by ICCs and SMCs with neglectable contribution by ENS, (2) self-organization ability of internal ICCs encapsuled by a thin layer of SMCs, (3) recurrent and stable contractions. Based on these characteristics, we designated this spheroid as “gut contractile organoid”.

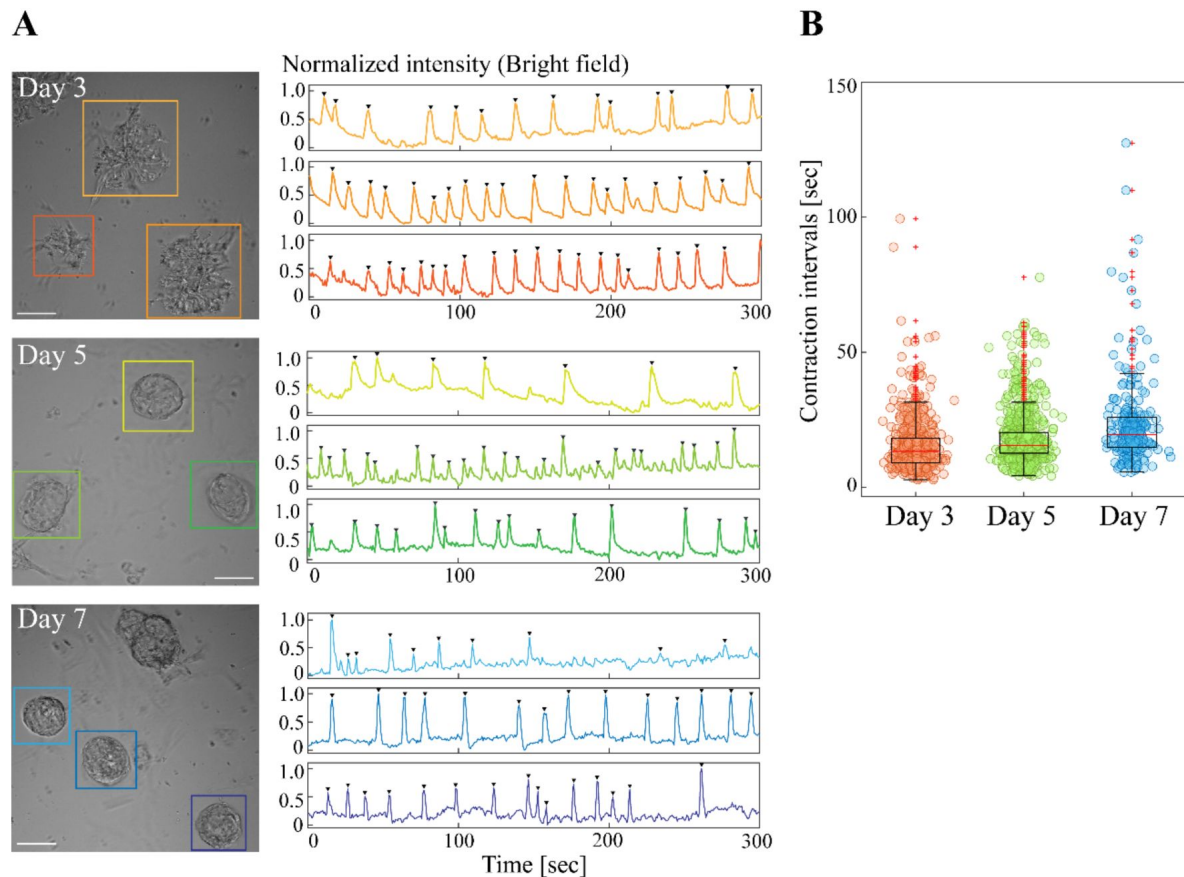


Figure 2.

Spheroids formed in Neurobasal medium and Matrigel exhibited reiterated contractions.

(A) Clusters/spheroids at days 3, 5, and 7 exhibited reiterated contractions. Graphs show normalized contraction intensities visualized using the Time Measurement function. Arrowheads indicate contraction peaks defined by a peak prominence > 0.25 and a peak width ≤ 10 seconds. **(B)** Contraction intervals in clusters/spheroids from day 3 to day 7. Each dot represents a single contraction interval. Median values: day 3, 13.3; day 5, 15.4; day 7, 19.6. Sample sizes: day 3, $n = 35$, peak count = 646; day 5, $n = 48$, peak count = 748; day 7, $n = 21$, peak count = 250. Scale bar: 50 μm (A).

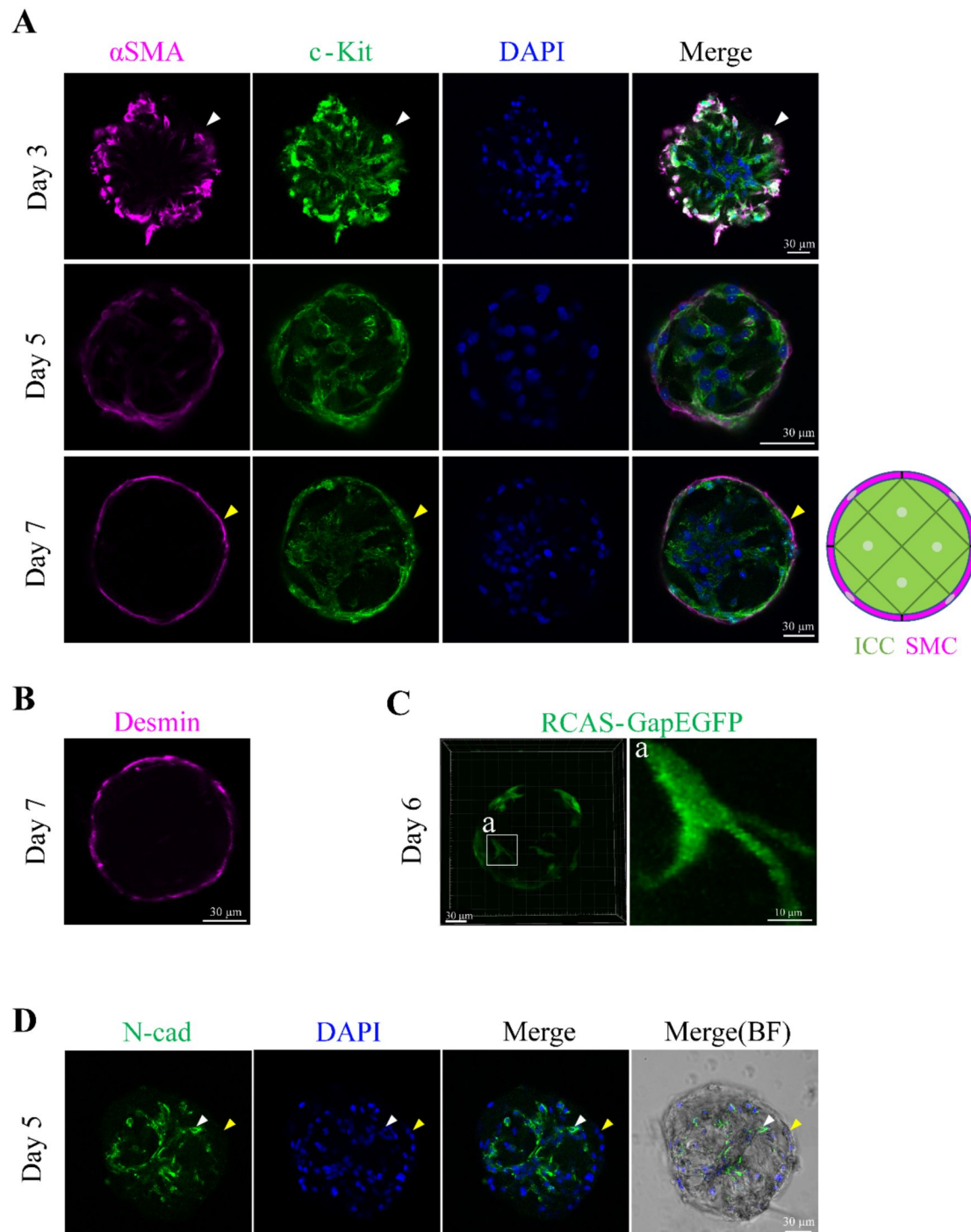


Figure 3.

Clusters/spheroids are composed of internally located ICCs and peripherally located SMCs.

(A) Co-staining with anti-c-Kit and anti- α SMA antibodies. White arrowheads indicate co-expression of c-Kit and α SMA at day 3. Yellow arrowheads indicate cells expressing α SMA but not c-Kit at day 7. A schematic diagram illustrates the spatial arrangement of cells within a day 7 spheroid (green: ICCs; magenta: SMCs). **(B)** Staining with anti-Desmin antibody, a marker for mature smooth muscle cells. **(C)** Cell morphology within the spheroid at day 6 visualized by RCAS-gapEGFP expression. **(D)** Staining of day 5 spheroids with anti-N-cadherin antibody. A white arrowhead indicates N-cadherin-positive cells, while a yellow arrowhead indicates N-cadherin-negative cells in the outer region of the spheroid. Scale bars: 30 μ m (A–D), 10 μ m (inset a in C).

Contraction-associated intracellular Ca^{2+} transients were coordinated between ICCs and SMCs in the gut contractile organoid

It has been reported that Ca^{2+} dynamics are important for pacemaker activity in ICCs and contractions of SMCs. Ca^{2+} flows into ICCs via voltage-dependent Ca^{2+} channels and propagates to SMCs, causing the gut muscle contractions (Baker et al., 2021). We therefore investigated the Ca^{2+} dynamics in our gut contractile organoids. Organoid-forming cells were infected with a RCAS-vector encoding GCaMP6s, a Ca^{2+} indicator that emits EGFP fluorescence in response to Ca^{2+} influx, and mRuby3 as a reporter (Fig. 4A). GCaMP6s-organoids were subjected to time-lapse imaging analyses by confocal microscopy. As expected, the oscillatory rhythm of Ca^{2+} transients as a whole organoid was highly concomitant with that of contractions (Fig. 4B, Movie 4).

Taking advantage of the spatial segregation between ICCs (internal) and SMCs (peripheral) (Fig. 3A, B), we compared the Ca^{2+} oscillatory rhythm among ICCs and SMCs (homotypically) and between ICC-SMC (heterotypically). We set up 3 regions of interests (ROIs) in the SMC layer with one ROI corresponding to one cell, and captured the Ca^{2+} transients. The three ROIs exhibited a synchronous pattern of Ca^{2+} oscillations (Fig. 4C, SMCs). Similarly, Ca^{2+} oscillations in three ROIs in the central region (ICCs) were synchronous (Fig. 4C, ICCs). These data highlight active communications taking place intercellularly within SMCs and ICCs, respectively. We further compared Ca^{2+} oscillations between ICC-SMC by setting up one ROI in each of ICC and SMC. Again, the Ca^{2+} rhythm was synchronized between these heterotypic cell types (Fig. 4C, ICC/SMC). With a deeper scrutinization, a peak of Ca^{2+} transient in ICC preceded that in SMC with a time lag (also called latency) of 700 msec in average (Fig. 4C, ICC/SMC; Fig. 4D, 117 peaks for 14 organoids), suggestive of a signal propagation from ICC to SMC consistent with previous reports (Baker et al., 2021).

Gap junction plays a role in ICC-to-SMC signaling

Although a series of gut motility studies have proposed an importance of gap junctions, rigid evidence has been limited due to a lack of experimental model. Our gut contractile organoid should prove useful for clarifying the roles of gap junctions in the synchronous motility, since intercellular synchronization was observed between/among identifiable cells (ICCs and SMCs) in live as shown above. We performed pharmacological assessments using gap junction inhibitors including carbenoxolone (CBX) and 18 β -glycyrrhetic acid (18 β -GA; Chevalier, 2018; Takeda et al., 2005), two of the most widely used pharmacological inhibitors for gap junction studies.

Following inhibitor administration into day 7 culture medium, organoids were allowed to rest for 30 min to exclude possible effects by the administration, for example, turbulence of the medium. Contrary to our expectation, none of 100 μM concentrations of CBX or 18 β -GA showed detectable effects on the recurring contractions of an organoid with frequency/interval comparable to the control (Fig. S3A and B, Movie 3). These undetectable effects were not due to insufficient penetrance of the drugs into the organoid, since the drug administration at day 3, when organoid-forming cells were still at 2D in culture, also yielded undetectable effects assessed at day 7 (Fig. S3C). The inhibiting activity of the drugs used here was verified using embryonic heart cultured cells (Fig. S3D).

Such unchanged patterns were also observed for the synchronization/ coordination of Ca^{2+} transients either in ICCs or SMCs in experiments conducted in a way similar to Fig. 4 (Fig. 5A, B, Movie 5). Same cells in an organoid were tracked for their Ca^{2+} transients before and after the inhibitor administration. With a closer look, however, while overall synchronization was retained between ICCs and SMCs, the preceding peak of Ca^{2+} transient in ICC was abolished (Fig.

5C, D [↗](#)). Collectively, while the contribution by gap junction to the periodic contraction and intercellular synchronization in the Day 7 organoid is relatively limited, the ICC-to-SMC signals require gap junction-mediated communications, at least partly.

Blebbistatin and Nifedipine ceased organoidal contractions and oscillatory patterns of Ca^{2+} transients

Toward searching for factors that regulate the coordination between/among ICCs and SMCs, we tested blebbistatin, a specific inhibitor of myosin II, which was expected to cease organoidal contractions. Experimental procedures were similar to those for gap junction inhibitors. We found that blebbistatin ceased periodic contractions of organoids in a concentration-dependent manner: while intervals were shorter with smaller amplitude at 5 μM , contractions were ceased completely at 10 μM (**Fig. 6A, B** [↗](#), Movie 6). The 10 μM -treated specimens resumed contractions following medium-wash out, showing that the organoids were alive (**Fig. 6B** [↗](#)).

We examined Ca^{2+} transients in these contraction-inhibited organoids. Markedly, periodic Ca^{2+} transients were extinguished not only in SMCs but also in ICCs, yielding no/little synchronous Ca^{2+} patterns among and between ICCs and SMCs (**Fig. 6C-F** [↗](#)).

Although a possible direct inhibition of non-muscle myosin II in ICCs cannot be excluded, these findings raised an interesting possibility that the contractility feeds back to ICCs to generate/maintain their periodic rhythm.

This notion was further corroborated by similar experiments using Nifedipine, a blocker of L-type Ca^{2+} channel known to function in gut SMCs (Chevalier et al., 2024; Der et al., 2000). Organoidal contractions were completely ceased by 1 μM Nifedipine, and resumed after washout, at least partly (**Fig. 7A, B** [↗](#)). In Nifedipine-treated organoids, Ca^{2+} transients (GCaMP) in ICCs were erased, which resumed following washout (**Fig. 7C** [↗](#)).

Inter-organoidal coordination was mediated by SMCs

During analyses with our novel organoids, we noticed that they easily fuse to each other, suggesting that organoids grow by progressive fusion (**Fig. 8A** [↗](#), Movie 7). This also raised the possibility that the synchronous Ca^{2+} transients among organoid-constituting cells (**Fig. 4C** [↗](#)) might be a consequence of phasic coordination upon the fusion of multiple organoids that had shown different oscillatory phases. To test this possibility, we transferred two organoids in a petri dish, and allowed them to fuse. Before the fusion, Ca^{2+} oscillatory rhythm was indeed out of phase in the two organoids (**Fig. 8B** [↗](#), Movie 8). Markedly, upon fusion by 24 hours, their rhythm became in phase/synchronous (**Fig. 8B** [↗](#), Movie 8). Intriguingly, they underwent a “pause” of oscillation during fusion (see Discussion).

Since we noticed that cellular protrusions were often observed around the time of organoidal fusion, we reasoned that these cellular processes would mediate the fusion and its subsequent synchronization of Ca^{2+} transients (**Fig. 8D** [↗](#)). To test this, we developed a 3-well hydrogel with narrow channels connecting the wells (**Fig. 9A** [↗](#)). Organoids were placed separately in each well, which prevented the fusion of organoidal bodies, but allowed extension of cellular processes through the narrow channels and contact with each other (**Fig. 9B** [↗](#)). After 72 hours of placement of organoids, cellular processes as well as several cell bodies were present within the channel, and Ca^{2+} transients became synchronized among the three organoids (**Fig. 9C** [↗](#), Movie 9). However, with careful examination, we also noticed that organoid-derived cells crawled out from the wells to cover the top surface of the hydrogel, connecting the three organoids (**Fig. 9D** [↗](#)). Thus, another possibility was raised that these crawled out cells would mediate the inter-organoidal synchronization.

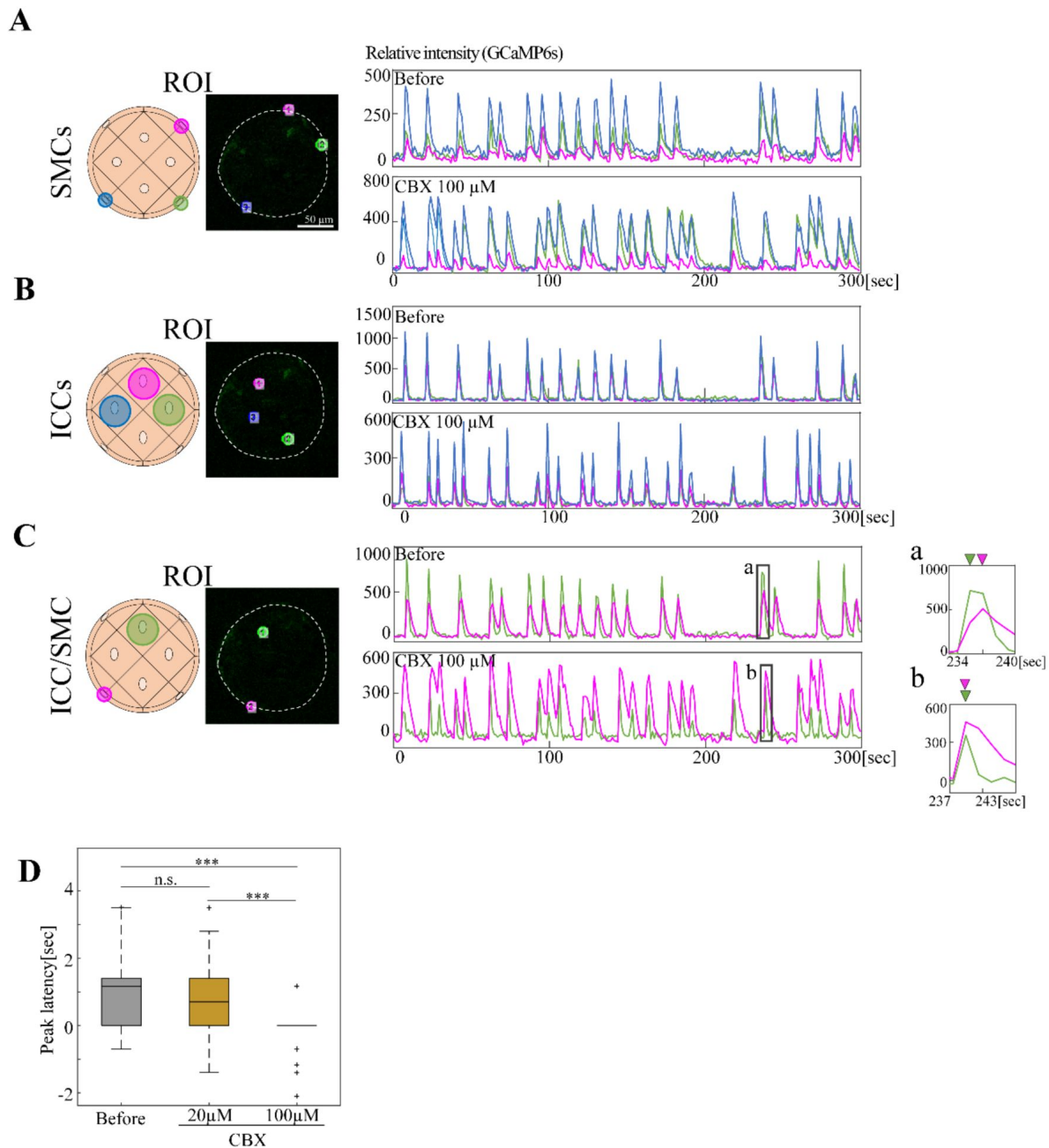


Figure 5.

Gap junction inhibitor exerted limited effects on the synchronization of Ca^{2+} dynamics.

Ca^{2+} synchronization among two or three ROIs in GCaMP6s-expressing organoids was evaluated before and after treatment with 100 μ M CBX. The synchronization was unaffected between **(A)** SMC-SMC and **(B)** ICC-ICC, but was partially affected between **(C)** ICC-SMC. Magnified views **(a, b)** highlight that the preceding Ca^{2+} peak in ICC (green) observed before CBX treatment **(a)** was abolished after treatment **(b)**. **(D)** Peak latency between ICC and SMC. Median values: Before, 1.167; CBX 20 μ M, 0.701; CBX 100 μ M, 0.0. Statistical significance was assessed using Welch's t-test. *** $p < 0.001$, n.s., $p \geq 0.05$ ($p = 0.27$ for Before vs. CBX 20 μ M). Sample sizes: Before: $n = 10$, peak count = 50; CBX 20 μ M: $n = 5$, peak count = 34; 100 μ M: $n = 5$, peak count = 51, Scale bar: 50 μ m (A).

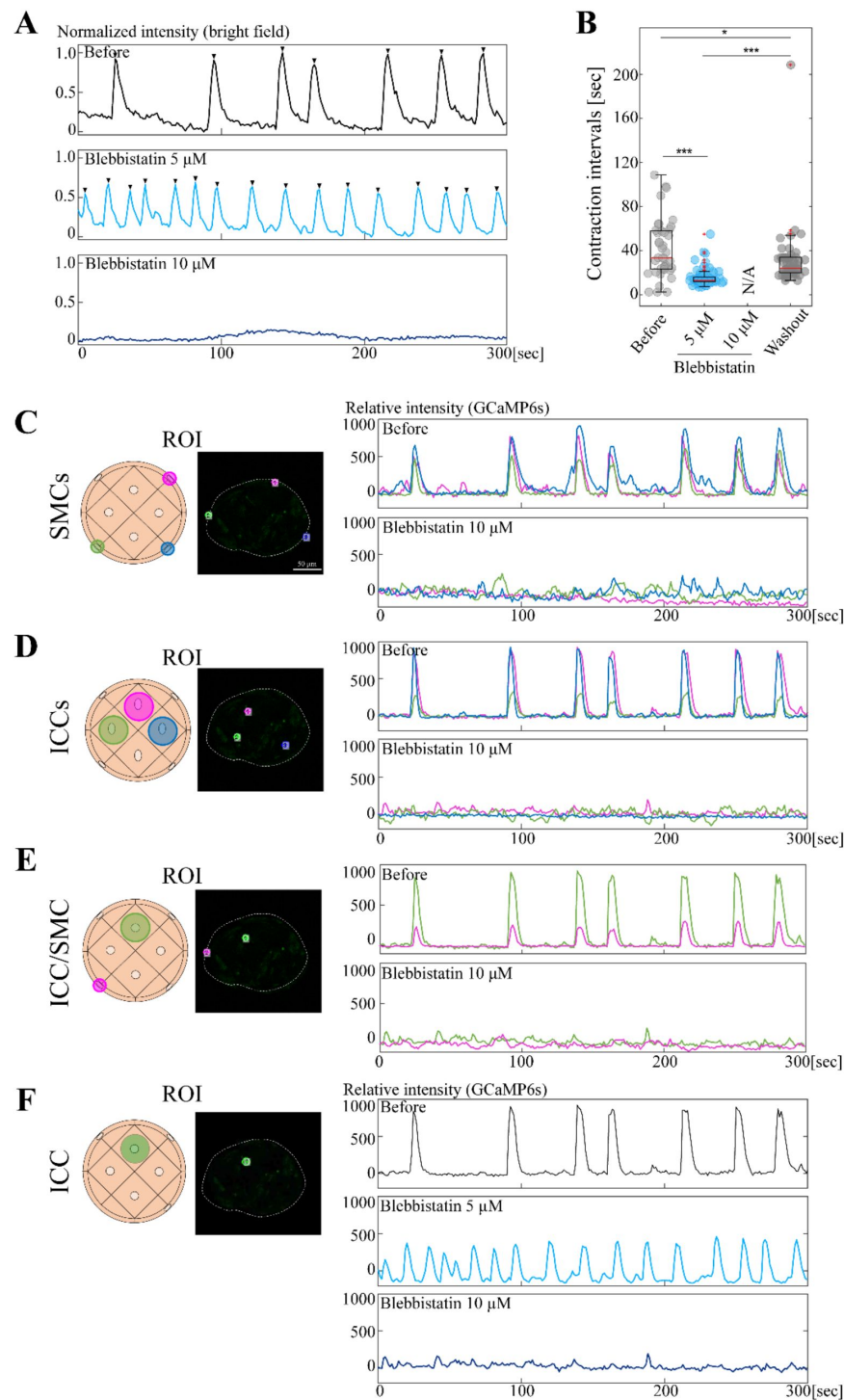


Figure 6.

The organoidal contraction is important for Ca^{2+} dynamics in ICCs.

GCaMP6s-expressing organoids were cultured with Blebbistatin. **(A)** Organoidal contractions were extinguished at 10 μ M. **(B)** Contraction intervals before and after Blebbistatin, and upon washout. Median values: Before, 33.46; Blebbistatin 5 μ M, 12.86; 10 μ M, N/A; Washout, 23.95. Statistical significance was assessed using Welch's t-test. *** $p < 0.001$, * $p < 0.05$. Sample size: $n = 3$. **(C-E)** Comparison of Ca^{2+} dynamics in SMC-SMC, ICC-ICC, and ICC-SMC. Three or two ROIs were assessed before and after administrations of 10 μ M Blebbistatin. **(F)** Ca^{2+} transients in a single ICC at 0 μ M (before), 5 μ M, and 10 μ M. Scale bar: 50 μ m (B).

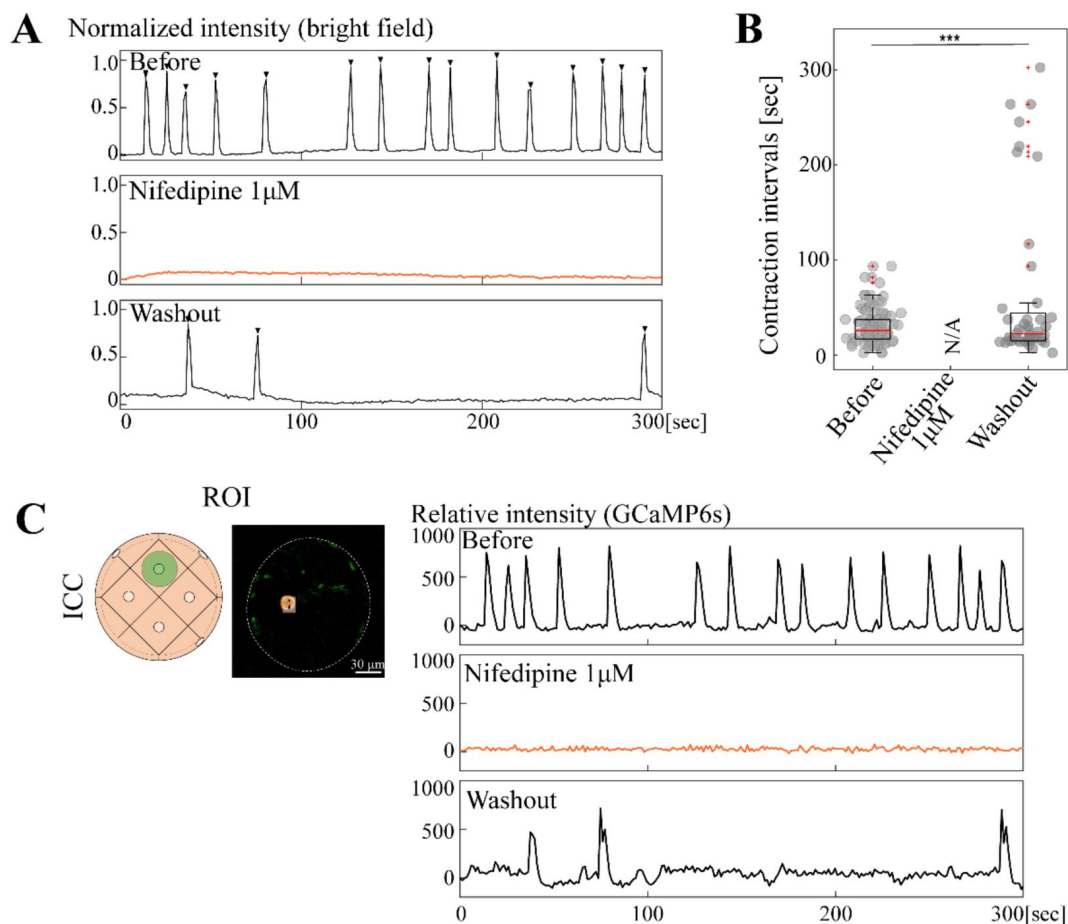


Figure 7.

Nifedipine ceased organoidal contractions and Ca^{2+} activities in ICCs.

Similar experiment to [Fig. 6](#). **(A)** Organoidal contractions were extinguished at 1 μ M. **(B)** Contraction intervals before and after administration, and upon washout. Median values: Before, 25.7; Nifedipine 1 μ M, N/A; Washout, 22.2. Statistical significance was assessed using Welch's t-test. $**p < 0.01$. Sample size: $n = 4$. Peak counts: Before, 101; Nifedipine 1 μ M, N/A; Washout, 46. **(C)** Ca^{2+} transients in a single ICC before and after administration of 1 μ M nifedipine, and following washout. Scale bar: 50 μ m (B).

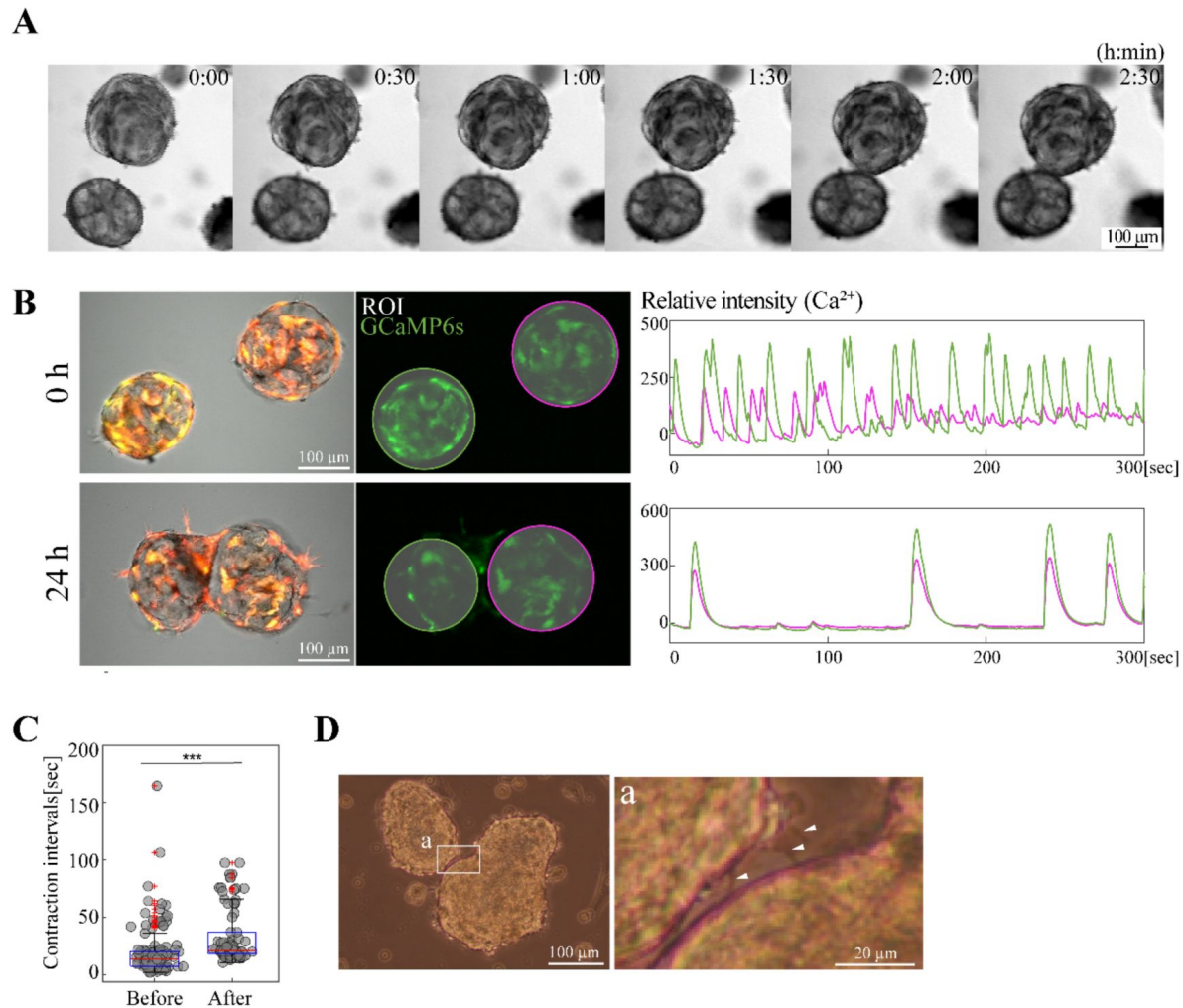


Figure 8.

Ca^{2+} transients in multiple organoids undergo synchronization upon organoidal fusion.

(A) Time-lapse imaging of organoidal fusion. **(B)** When two organoids that originally displayed independent Ca^{2+} rhythm fused to each other, their rhythm became synchronized after fusion (24 h). **(C)** Contraction intervals before and after fusion. Median values: before fusion, 13.65; after fusion, 21.04. Statistical significance was assessed using Welch's t-test. *** $p < 0.001$. Sample size: $n = 3$ pairs. Peak counts: before, 138; after, 90. **(D)** Cellular protrusions between two neighboring organoids. White arrowheads show three protrusions from the left organoid. Scale bars: 100 μ m (A, B, D), 20 μ m (inset (a) in (D)).

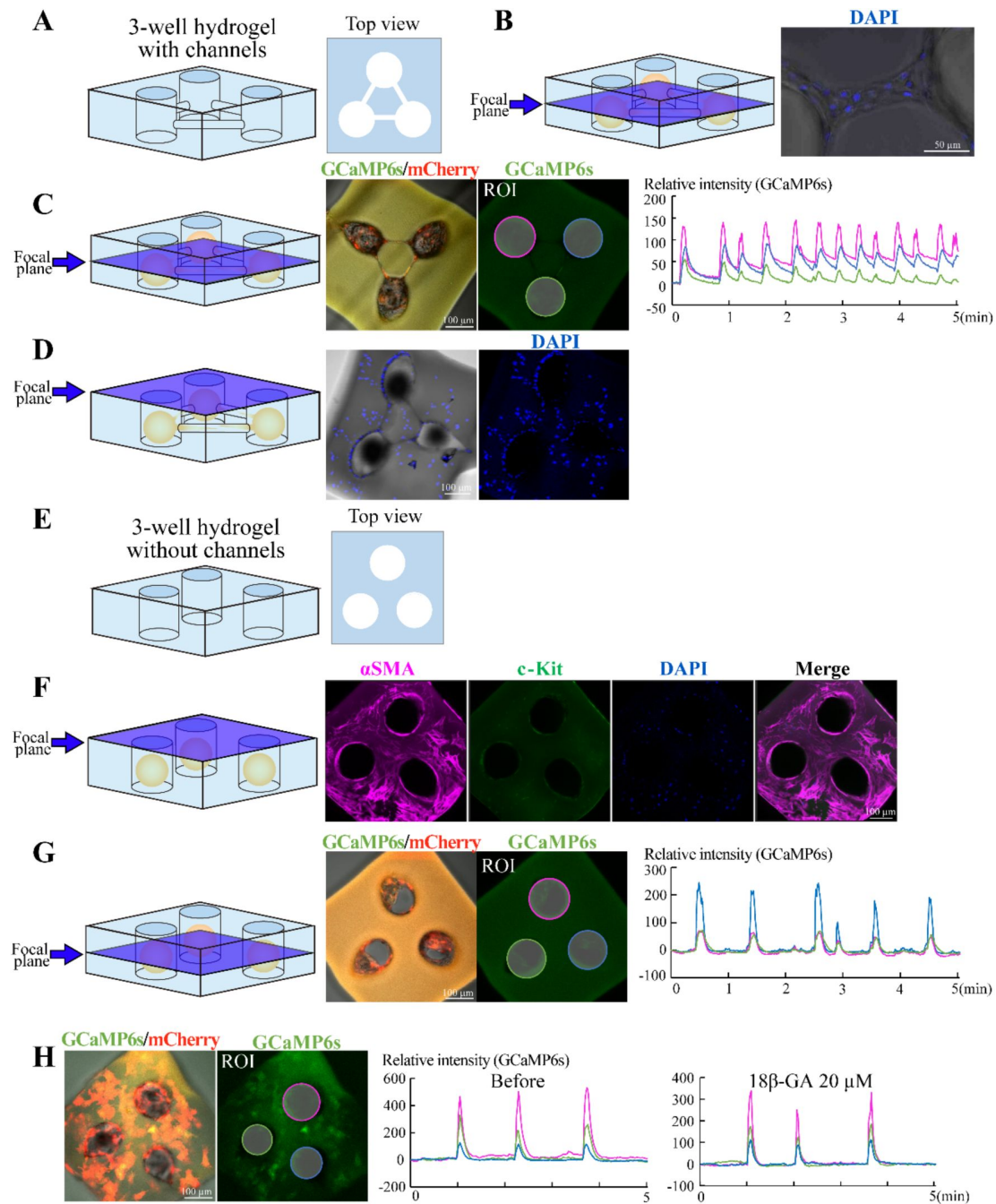


Figure 9.

Smooth muscle cells mediate Ca^{2+} synchronization between organoids.

(A) Diagram of a 3-well hydrogel in which one organoid was placed per well. The three wells were connected with narrow channels, and this gel mold does not allow organoidal bodies to fuse to each other, but allow them to extend/migrate protrusions/cells through the channel (B). (C) After 3 days, the three organoids displayed synchronization of Ca^{2+} dynamics. (D) Some organoid-derived cells crawled out from the wells, and covered the top surface of the hydrogel, resulting in bridging the three unfused organoids. (E) Diagram of a 3-well hydrogel *without* channels. (F) The surface-covering cells were identified as SMCs (αSMA -positive, $c\text{-Kit}$ -negative). (G) In the hydrogel without channels but with surface-covered SMCs, Ca^{2+} dynamics in the three organoids were synchronized. (H) The Ca^{2+} synchronization shown in (G) was not altered by treatment with $18\beta\text{-GA}$. Blue planes indicate focal planes. Scale bars: 50 μm (B); 100 μm (C, D, F, G, H).

To test this possibility, we prepared a similar 3-well hydrogel, but in this case the three wells were disconnected (no channels) so that organoid-derived cellular processes were not able to connect each other (**Fig. 9E**). By 3 days of culture, the top surface of the hydrogel was indeed covered by cells in a similar way to **Fig. 9D**. These cells were positive for α SMA but negative for c-Kit (**Fig. 9F**), showing that they were SMCs that were somehow detached and crawled out from the peripheral layer of their “host” organoids. Importantly, coincided with the top-coverage by the SMCs, the three organoids in the disconnected wells displayed synchronized Ca^{2+} transients, highlighting the role of SMCs in mediating coordination between organoids (**Fig. 9G**, Movie 10). Gap junction inhibitor yielded no/little effects on the Ca^{2+} transient coordination (**Fig. 9H**, Movie 11).

Discussion

We have developed a novel gut contractile organoid, which displays several unique characteristics: 1) it undergoes recurrent contractions, 2) differentiation states of ICCs (c-Kit⁺ α SMA⁻) and SMCs (c-Kit⁺ α SMA⁺) are maintained at least until day 7 in the organoid, 3) the organoid is composed essentially of two types of cells, ICCs and SMCs, with few ENS cells, if any, 4) ICCs (internal) and SMCs (peripheral) can be distinguished for their localization in a living organoid, allowing 5) GCaMP-visualization of Ca^{2+} transients and assessments of cell interactions between and among ICCs and SMCs. These characteristics circumvent, at least partly, obstacles that have hampered analyses in the research of gut peristalsis, such as unstable differentiation state of ICCs and SMCs in cultures, and difficulties in identifying these cells in living preparations. In studies of gut movements, how ICCs generate/maintain their periodic rhythm and how ICCs and SMCs interact with each other have been long-standing questions, and our organoids offer powerful advantages to address these fundamental questions and to understand the cellular mechanisms underlying the gut contractions/peristaltic motility at least in the embryonic gut.

Contrasting with many cases in organoid studies that aim at a *maximum* recapitulation of the intact organ, our gut contractile organoid is composed of a (probably) *minimum* number of cell types that suffice the generation and/or maintenance of rhythmic contractions, allowing high-resolution analyses at the cellular level. For example, adding ENS components to our organoid would allow the clarification of the role of ENS in the gut contraction. While the majority of internal cells in the organoid are ICCs (c-Kit⁺ α SMA⁻), the possibility cannot be excluded that platelet-derived growth factor receptor- α positive (PDGFR α ⁺/c-Kit⁺ α SMA⁻) cells, another type of interstitial cells (fibroblast-like cell) known to mediate neural activity to SMCs in the mouse gut (Sanders et al., 2024; Sanders et al., 2016), are included in our organoid. Available antibody against the chicken PDGFR α protein is awaited.

Coordinated Ca^{2+} transients in ICC/SMC populations in the gut contractile organoid

Measurement and quantification analyses with ICCs and SMCs that are identifiable in the living organoid revealed exquisite coordination of Ca^{2+} transients/oscillation homotypically in both ICC-ICC and SMC-SMC combinations, and heterotypically between SMC-ICC (**Fig. 4C**). Notably, a peak of Ca^{2+} transient in ICCs precedes that of SMCs with a time lag (also called latency) of 700 msec, implying a signaling from ICC to SMC. These observations are consistent with previous studies using ICC- and SMC-specific transgenic mice that expressed GCaMP and RCaMP, respectively (Baker et al., 2021). In that study, the authors assessed Ca^{2+} transients in submucosal ICCs (ICC-SM) and compared them with those in their adjacent circular muscles, and showed that the rise of GCaMP signal (ICC) preceded that of RCaMP (SMC) with a latency of 56 $\pm$ 14 msec. With these observations, they concluded that ICC-SM sends signals to its adjacent SMC. In our study, time-

lapse imaging was mostly performed with 700 msec intervals. Further studies with shorter intervals are awaited to know whether the latency time would be shorter than 700 msec in average in our organoids.

Contribution of gap junction to ICC-to-SMC signaling

Effects by the block of gap junction by CBX or 18β -GA were relatively limited in our organoid assays: organoidal contraction rhythm and synchronous patterns of Ca^{2+} transients remained unchanged between SMC-SMC and ICC-ICC. In contrast, signaling from ICCs to SMCs was affected, in which the preceding peak of Ca^{2+} transient in ICCs was abolished. This effect was seen by CBX but not by 18β -GA (**Fig. 5C** [↗](#), S3E). Further studies are required to clarify which connexin(s) plays a role in the generation of latency. The contribution of gap junction to the ICC-to-SMC signaling was previously reported in mouse gut acting from intramuscular ICCs (ICC-IM) to their adjacent circular smooth muscles. However, interpretation of the role of gap junction in ICC-SMC interactions, in general, has been under big debate. Some studies reported that CBX or 18β -GA failed to inhibit these interactions or peristaltic motilities ([Komuro et al., 1996](#) [↗](#); [Rohr et al., 1998](#) [↗](#); [Schultz et al., 2003](#) [↗](#)), or that electron microscopy did not detect structures of gap junction in longitudinal muscles ([Cousins et al., 2003](#) [↗](#); [Daniel & Wang, 1999](#) [↗](#); [Gabella & Blundell, 1981](#) [↗](#)). Contribution of gap junctions to the gut motility appears to be highly variable in different regions of gastrointestinal tract, for example, stomach versus colon ([Iino et al., 2007](#) [↗](#); [Yang et al., 2012](#) [↗](#)). And a detection of gap junction structures or mRNA/protein of connexins does not necessarily mean that the gap junction is functional. Dominant negative form of gap junction might be useful. Currently, ICC- or SMC-specific gene manipulations in our organoid are not available, and further studies are needed. One possibility is that while gap junctions are important for the onset of rhythm coordination, once the rhythm is established, other mechanisms might be employed, for example, mechanical feedback from muscles to ICCs (see below). Similar notion has recently been shown for coordinated Ca^{2+} signaling in cardiac muscle cells ([Fukui et al., 2021](#) [↗](#)).

Possible feedback from SMC's contractility to ICC's oscillatory rhythm

Blebbistatin extinguished the contraction of the organoids, which was concomitant with abrogation of Ca^{2+} transients in ICCs (**Fig. 6** [↗](#)). Since it has been reported that thick myosin fibers necessary for the contraction are found only in SMCs but not in ICCs ([Gherghiceanu & Popescu, 2005](#) [↗](#); [Rumessen & Thuneberg, 1991](#) [↗](#), [1996](#) [↗](#); [Sun et al., 2006](#) [↗](#)), it is likely that ICCs do not have contractile ability. Our observations therefore raise the possibility that ICC's pace-making activity requires mechanical feedback from contracting SMCs. This notion is also supported by additional finding that Nifedipine, an L-type Ca^{2+} channel blocker known to be expressed in SMCs ([Chevalier et al., 2024](#); [Der et al., 2000](#)), erases both organoidal contractions and Ca^{2+} transients in ICCs (**Fig. 7** [↗](#)). The possibility of the SMC-to-ICC signaling is further corroborated by other findings obtained in this study showing that SMCs mediate inter-organoidal rhythm coordination (**Fig. 9** [↗](#)) as more discussed below.

It is of note that the reciprocal interactions between pace-making cells and their effectors have been reported in studies of neural circuit establishment. Spontaneous activities emerging in motor neurons during peristaltic locomotion of larvae in *Drosophila* are regulated by feedback from their governing muscle's contractions, so that motor neurons get organized to display coordinated and stable oscillatory activities ([Zeng et al., 2021](#) [↗](#)). Thus, it is tempting to speculate that during early gut peristalsis, ICCs that have initiated their spontaneous activities receive feedback from their governing SMCs to generate more stable coordination of pace-making activity among ICCs. Indeed, we have previously reported that in the very early embryonic gut, origins of peristaltic wave (OPWs) are randomly distributed along the gut axis, but these unstable OPWs later become confined to specific sites displaying more stable and coordinated pace-making patterns ([Shikaya et al., 2022](#) [↗](#)).

Interactions between ICCs and SMCs in the gut contractile organoid

The organoid developed in this study was derived from the muscle layer (also called *tunica muscularis*) of chicken E15 hindgut devoid of mucosa and serosa, in which myenteric ICCs (ICC-MY), intramuscular ICCs (ICC-IM), and submucosal ICCs (ICC-SM) are localized in a way similar to those in mice, shown by staining with antibody against the chicken c-Kit protein (Yagasaki et al., 2022 [↗](#)). In the current study, c-Kit antibody-staining showed two types of cells in morphology in the gut contractile organoid: one is multipolar and N-cad-positive ICCs located centrally, and the other is ICCs that are thin in shape, N-cad negative, and lining beneath the most external layer of SMCs. Based on the knowledge obtained in studies with mammalian species that ICC-MY are multipolar whereas ICC-IM are bipolar and tightly associated with adjacent SMCs (Huizinga et al., 2011 [↗](#); Iino & Horiguchi, 2006 [↗](#)), it is conceivable in our organoids that the central cells are ICC-MY, and the peripherally lining ones are ICC-IM. In addition, it has been reported in mice that signaling from ICC-IM to SMCs is gap junction-dependent. Collectively, our observations that gap junction-dependent signal found between ICC (central ICC) and SMC in the organoid (Fig. 5C, D [↗](#)) could be interpreted as follows: ICC-MY (central) signals to ICC-IM (second-most peripheral), which in turn acts on the external SMCs mediated partly by gap junction. Such sequences of signaling (ICC-MY to ICC-IM to SMCs) have also been proposed in the intact gut in mammals, although rigid evidence has not been known. At present, direct comparison of Ca^{2+} transients between the ICC-IM-like cells and SMCs in our organoid was technically very difficult since these two cells were both thin and tightly associated each other.

The gut contractile organoid provides useful model and tool for studying phase coordination of oscillatory rhythm

To understand the gut peristaltic movements, which reiterate at specific sites along the gut axis (Shikaya et al., 2022 [↗](#)), deciphering the mechanisms underlying the coordination/synchronization of oscillators among multiple cells is critical. Exploiting the finding obtained in the current study that multiple organoids easily fuse each other in vitro, we have found that two organoids with different oscillatory rhythm eventually coordinate their phases upon the fusion (Fig. 8 [↗](#)). This suggests the ability of ICCs to adjust their rhythm to their neighbors. It is tempting to speculate that during “pausing time” of oscillation upon organoidal fusion (Fig. 8B [↗](#)), they might communicate each other to adjust to a unified rhythm.

During identification of the rhythm-adjustment mediators using the 3-well hydrogel (without connecting channels), SMCs that were unexpectedly crawled out from the organoids and covered the top surface of the hydrogel were able to mediate the rhythm coordination among organoids (Fig. 9 [↗](#)). Whether cellular thin protrusions connected with each other observed in the channel-connected 3-well hydrogel mediate the coordination remains undetermined. Nevertheless, our findings have revealed a novel role of SMCs in mediating rhythm coordination, and as discussed above, these support the notion of the SMC-to-ICC signaling, which is unprecedented. It is unlikely that gap junction plays a major role in such signaling since gap junction inhibitors yielded no detectable effects in SMC-mediated organoidal phase synchronization (Fig. 9H [↗](#)).

In summary, the gut contractile organoid developed in this study serves as a powerful model to study the establishment and maintenance of oscillatory rhythm (pace-making) and their coordination in the multicellular systems.

Materials and methods

Chicken embryos

Fertilized chicken eggs were obtained from the Yamagishi poultry farms (Wakayama, Japan) and Takeuchi Farm (Nara, Japan). Embryos were staged according to embryonic days. All animal experiments were conducted with the ethical approval of Kyoto University (#202110).

Culture preparation of hindgut-derived cells

A hindgut was dissected from E15 chicken embryos (**Fig. S1A** [↗](#)), and cut into small pieces. After treating with 25 U dispase (Fujifilm Wako, 383-02281) /phosphate buffer saline (PBS: 0.14 M NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄-12H₂O, 1.8 mM KH₂PO₄) at 38.5 °C for 40 min, serosa and intestinal epithelium were removed using forceps. The muscle layer was minced into smaller pieces and treated with 0.2 mg/ml collagenase (Fujifilm Wako, 038-22361) and 0.25 % trypsin/PBS at 37.0 °C for 30 min. The reaction was stopped with 1 % FBS/PBS followed by centrifugation at 800 rpm for 3 min. The pellet was resuspended and washed in PBS followed by centrifugation. They were suspended in culture medium, and 5.0×10^5 cells were plated on 14-mm diameter glass-bottom dishes (Matsunami, D11130H) which had been treated with undiluted Matrigel (Corning, 354248) at 38.5°C for 20 min. Poly-lysine and collagen-coated dishes were purchased (Matsunami, D11131H, D11134H). D-MEM /Ham's F-12 (Wako, 048-29785), Ham's F-12 (Wako, 087-08335) and Neurobasal medium (Gibco, 21103-049) with 1× B-27 supplement (Gibco, 17504044) and 0.5 mM L-glutamine were tested. After seeding, time-lapse images were obtained with CM20 (Evident) under 5% CO₂ and 38.5°C with 2-hour intervals.

Assessment of contraction in the spheroid/cluster and organoids

Time-lapse images were obtained using confocal microscopy (Nikon, A1R) under 5% CO₂ and 38.5°C. Region of interest (ROI) was set using the Time Measurement function in Nikon NIS Elements, and changes were detected in the mode “stDev Intensity” and exported as individual CSV files. Bright-field changes were normalized to values between 0 and 1. For drug experiments, the values in the control group were used for normalization.

Using MATLAB (MathWorks), the changes were plotted, and peaks with a minimum prominence of 0.25 and a maximum width of 10 seconds were detected as contractions of spheroid/cluster or organoid using the findpeaks function. To determine the contraction frequency, we analyzed intervals between contractions and displayed as box plots.

Immunocytochemistry

An organoid was fixed in acetic acid/ethanol (1:5) for 10 min at room temperature (RT), and washed in PBS for 10 min at RT. The specimens were permeabilized in 0.1% Tween-20 in PBS for 10 min at RT, followed by washing in PBS for 10 min at RT. After blocking with 1 % blocking reagent for 1 hour at RT, specimens were incubated overnight at 4 °C with dilution of 1:300 anti-c-Kit (Yagasaki et al., 2022 [↗](#)), 1:300 Tuj-1 (RSD, MAB1195), 1:400 anti-αSMA antibody (Sigma-Aldrich, A5228), anti-Desmin antibody (Novus Biologicals, NBP1-97707) and/or 1:200 anti-N-cadherin antibodies (TAKARA, M110) in 1% blocking reagent (Roche, 1096176)/PBS. Following three times washing in PBS for 10 min each at RT, specimens were incubated for 1.5 hr at RT with 1:500 anti-rabbit IgG(H+L)-Alexa 488-conjugated antibody (Donkey; Invitrogen, A21206), anti-mouse IgG_{2a}-Alexa 568-conjugated antibody (Goat; Invitrogen, A21134), anti-rat IgG (H+L)-Alexa 488-conjugated antibody (Goat; Invitrogen, A11006) and 1:2000 DAPI. After washing three times in PBS for 10 min at RT, fluorescent images were obtained using the Nikon A1R confocal microscope.

Plasmids

pAAV-hSynapsin1-GCaMP6s-P2A-mRuby3 was purchased from Addgene (112007). Full-length cDNA of GCaMP6s was PCR-amplified:

forward 5'- GCGTACCACTGTGGCATCGATGCCACCATGGGTTCTCA -3',

reverse 5'- GCCCGTACATCGCATCGATTACTTGTACAGCTCGT -3'.

The retroviral vector RCAS-EGFP was digested with ClaI to remove EGFP, into which a DNA fragment was inserted by In-Fusion HD Cloning Kit (TAKARA) to produce RCAS-GCaMP6s-P2A-mRuby3. RCAS-GapEGFP is as previously described ([Murai et al., 2015](#) [↗](#))

Preparation of retroviral vector particles

RCAS-GCaMP6s-P2A-mRuby3 was transfected into the chicken fibroblast line cells DF1 using Lipofectamin 2000 (Invitrogen). Transfected cells were cultured in 10 cm culture dish until confluent. The supernatant of transfected DF1 was collected for viral precipitation, from which retroviral particles were prepared using Retro-Concentin™ Virus Precipitation Solution (SBI, RV100A-1).

Intracellular Ca²⁺ imaging in the gut contractile organoid

On day 2 or 3 of cell culture, a 10 mg/ml polybrene solution (final concentration: 4 µg/ml; Nacalai, 12996-81) and 20 µl of Opti-MEM containing the aforementioned viral particles were added to the culture medium to transfect GCaMP6s into organoid-forming cells. The culture medium was replaced with fresh medium on day 5, and Ca²⁺ imaging was performed on day 7. Time-lapse images were acquired using a confocal microscope (Nikon, A1R) at intervals of either 700 or 450 msec. Fluorescence intensity from each region of interest (ROI) was exported into an Excel file using the time-measurement function of the NIS-Elements software (Nikon). Fluorescence intensity traces were plotted using the Excel data, with the start of measurement set as time zero. For peak latency analysis, calcium transients in ICCs and SMCs were identified using the “findpeaks” function in MATLAB (MathWorks). Temporal differences between peak signals in ICC and SMC were calculated to show the latency.

Drug administration

Carbenoxolone (CBX; nacalai tesque, 32775-51) /H₂O, 18beta-Glycyrrhetic acid (18β-GA; abcam, ab142579)/DMSO, (-)-Blebbistatin (FUJIFILM Wako, 021-17041) and Nifedipine (FUJIFILM Wako, 141-05783) were prepared. Time-lapse images were acquired for a single organoid before and after the drug administration. Following the drug adding, organoids were allowed to rest for 30 min to avoid possible effect of turbulence of the medium, and time-lapse images were taken for 5 min.

Three-well hydrogel fabrication

The photoinitiator P2CK was synthesized as previously reported ([Li et al., 2013](#) [↗](#)).

The target product was verified using proton NMR and Bruker's TopSpin software. Gelatin-Norbornene was synthesized based on previous report ([Van Hoorick et al., 2018](#) [↗](#)). The final reaction mixture was transferred to a dialysis tube (5-6 kDa; Spectra Por, cat. no. 132680T) for dialysis against pure water at 40°C for 3 days. After dialysis, the solution's pH was carefully adjusted to 8.0.

A piece of 3-well hydrogel was fabricated on a glass bottom dish (Matsunami) from 100 μ L of a solution containing 20 wt % Gelatin-Norbornene, 2 mM of the photoinitiator P2CK, and 20 mM of crosslinker (DTT) (TCI, #D1071) using a two-photon microscope with controllable laser power in 3D space according to the voxel file input (Olympus, in house customized). The voxel files specifying the 3D shape of the hydrogel were designed with Fusion 360 software. (Autodesk). Organoids were transferred to the hydrogel using a glass capillary.

Culture of embryonic heart-derived cells

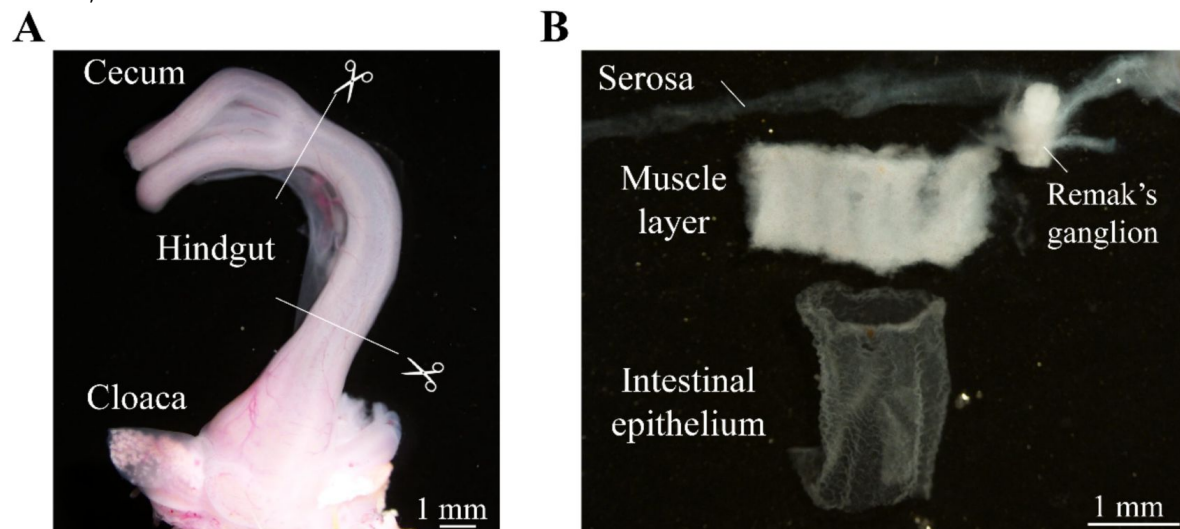
A heart was dissected from E10 chicken embryo, and cut into small pieces. The tissue fragments were treated with 0.125% trypsin in PBS at 37.0 °C for 30 min. The reaction was terminated by adding 10 % FBS in D-MEM/Ham's F-12, followed by centrifugation at 800 rpm for 3 min. Pellet was resuspended in D-MEM/Ham's F-12 supplemented with 10% FBS, and the cells were plated in poly-L-lysine-coated glass-bottom dishes (Matsunami).

On day 1, time-lapse images were acquired using confocal microscopy under 5% CO₂ at 38.5 °C, with 700 msec intervals. Cell contractions were detected using the “stDev Intensity” mode in the Time Measurement function of Nikon NIS-Elements.

Supplementary figures

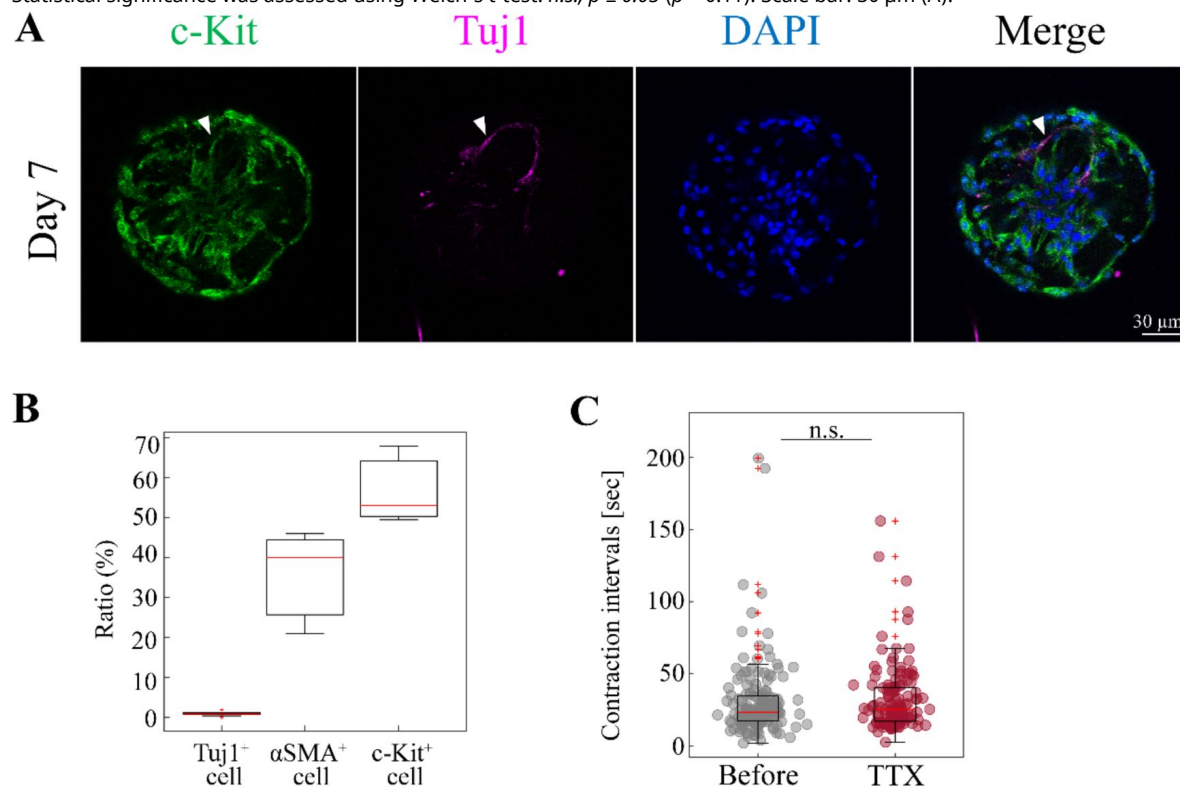
Supplementary Figure S1

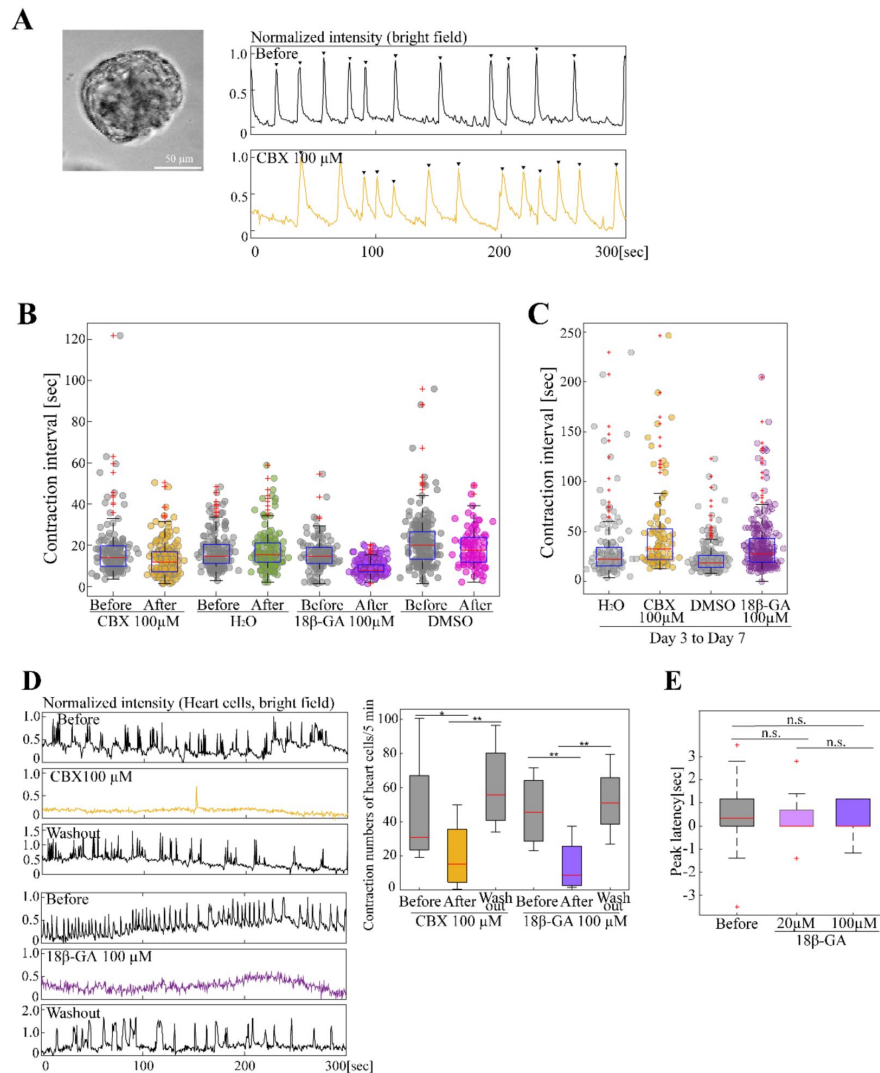
(A) Chicken embryonic hindgut at E15. It was dissected from the bottom of the cecum to the front of the cloaca (white lines). (B) Three layers of E15 hindgut: serosa, muscle layer, intestinal epithelium. Remak's ganglion was also removed. Scale bars: 1 mm in A, B



Supplementary Figure S2

(A) Co-staining of Day 7 organoids with anti-c-Kit- and Tuj1 antibodies. White arrowhead shows a Tuj1⁺ cell. (B) Representation of Tuj1-positive, α SMA-single positive, c-Kit-single positive cells in an organoid at day 7. Median values: Tuj1⁺, 0.92; α SMA⁺, 40; c-Kit⁺, 53. Sample size: Tuj1⁺, $n = 12$; α SMA⁺, c-Kit⁺, $n = 3$ (C) Contraction intervals before and after administration of TTX. Median values: Before, 23.4; TTX, 25.35. Sample size: $n = 9$. Peak counts: Before, 166; TTX, 148. Statistical significance was assessed using Welch's t-test. *n.s.*, $p \geq 0.05$ ($p = 0.11$). Scale bar: 30 μ m (A).





Supplementary Figure S3

Effects of gap junction inhibitors on organoidal contractions and ICC-SMC latency.

(A) Organoids at day 7 exhibited rhythmic contractions before and after administration of 100 μ M CBX. Graphs show normalized contraction intensities visualized using the Time Measurement function. Arrowheads indicate contraction peaks defined by a peak prominence > 0.25 and a peak width ≤ 10 seconds. **(B)** Contraction intervals before and after treatment with CBX (100 μ M), H₂O (vehicle control for CBX), 18 β -GA (100 μ M), and DMSO (vehicle control for 18 β -GA). Median values: CBX, 14.0 (before), 11.9 (after); H₂O, 14.7 (before), 15.4 (after); 18 β -GA, 14.7 (before), 7.7 (after); DMSO, 19.95 (before), 17.5 (after). Sample sizes and peak counts: CBX, $n = 9$ (before), $n = 9$ (after), peak count = 150 (before), 196 (after); H₂O, $n = 10$ (before), $n = 10$ (after), peak count = 173 (before), 171 (after); 18 β -GA, $n = 7$ (before), $n = 8$ (after), peak count = 129; DMSO, $n = 7$ (before), $n = 6$ (after), peak count = 193 (before), 92 (after). **(C)** Contraction intervals at day 7 in organoids cultured with CBX, H₂O, 18 β -GA, and DMSO from day 3 to day 7. Median values: H₂O, 22.75; CBX, 32.5; DMSO, 18.85; 18 β -GA, 27.95. Sample sizes and peak counts: H₂O, $n = 9$, peak count = 136; CBX, $n = 11$, peak count = 135; DMSO, $n = 9$, peak count = 209; 18 β -GA, $n = 18$, peak count = 268. **(D)** Peak latency of Ca²⁺ transients between ICC and SMC before and after administration of 18 β -GA (20 μ M, 100 μ M). Statistical significance was assessed using Welch's t-test. n.s., $p = 0.27$ for Before vs. 18 β -GA 20 μ M; $p = 0.25$ for Before vs. 18 β -GA 100 μ M; $p = 0.46$ for 18 β -GA 20 μ M vs. 100 μ M. Sample sizes and peak counts: Before, $n = 5$, peak count = 62; 18 β -GA 20 μ M, $n = 3$, peak count = 52; 18 β -GA 100 μ M, $n = 2$, peak count = 57. **(E)** Contractions of cultured embryonic heart cells. Graphs show normalized contraction intensities using the Time Measurement function. The box plot shows the number of contractions per 5 minutes. Median values: CBX, 30.75 (before), 15 (CBX 100 μ M), 55.75 (washout); 18 β -GA: 45.5 (before), 8.5 (18 β -GA 100 μ M), 51 (washout). Statistical significance was assessed using Welch's t-test. $**p < 0.01$, $*p < 0.05$; n.s., $p \geq 0.05$ ($p = 0.13$ for Before vs. washout of CBX; $p = 0.26$ for Before vs. washout of 18 β -GA). Sample sizes: CBX, $n = 4$; 18 β -GA, $n = 4$. Scale bar: 50 μ m (A).

Acknowledgements

We thank Dr. Scott Gilbert for careful reading of the manuscript and discussion. We also thank National BioResource Project (Chicken-Quail, Nagoya University) for their technical help. This work was supported by JSPS KAKENHI Grant Numbers; 23H04933, 20H03259, 20K20520, 20K21425 for Y. T., and 21K06198, 23H04702 for M. I., and FY 2022 Kusunoki 125 of Kyoto University 125th Anniversary Fund for M. I., and Ginpu Funds (Kyoto University) and incu-be fund (Leave a Nest Co., Ltd.) for R. Y.. R. Y. is an ex-fellow of JSPS.

Additional files

Movie 1. Long-term time-lapse imaging after seeding. Time lapse was taken every 2 hours for a total of 100 hours. This video corresponds to Fig. 1B. [🔗](#)

Movie 2. Periodic contractions of the cluster/spheroid on day 3, 5 and 7. Time-lapse images were taken with 700 msec intervals for 5 min. This video corresponds to Fig. 2A. [🔗](#)

Movie 3. Periodic contractions of a day 7 organoid with CBX. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. S3. [🔗](#)

Movie 4. Ca^{2+} dynamics in day 7 gut contractile organoid are concomitant with its contractions. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. 4. [🔗](#)

Movie 5. Ca^{2+} dynamics in a gut contractile organoid with CBX. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. 5. [🔗](#)

Movie 6. Ca^{2+} dynamics in day 7 gut contractile organoid with Blebbistatin. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. 6. [🔗](#)

Movie 7. Live imaging during fusion of multiple organoids. Time-lapse images were obtained every 10 minutes for 4 hours. This video corresponds to Fig. 8A. [🔗](#)

Movie 8. Ca^{2+} transients in two gut contractile organoids before and after fusion. Time-lapse images were obtained with 450 msec intervals for 5 min. This video corresponds to Fig. 8B. [🔗](#)

Movie 9. Ca^{2+} transients in three gut contractile organoids in 3-well hydrogel with channels. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. 9C. [🔗](#)

Movie 10. Ca^{2+} transients in three gut contractile organoids in 3-well hydrogel without channels. Time-lapse images were obtained with 450 msec intervals for 5 min. This video corresponds to Fig. 9G. [🔗](#)

Movie 11. Similar assay to movie 10 with 18β -GA administration. Time-lapse images were obtained with 700 msec intervals for 5 min. This video corresponds to Fig. 9H. [🔗](#)

References

- Baker S. A., Leigh W. A., Del Valle G., De Yturriaga I. F., Ward S. M., Cobine C. A., Drumm B. T., Sanders K. M. (2021) **Ca(2+) signaling driving pacemaker activity in submucosal interstitial cells of Cajal in the murine colon** *eLife* **10**:e64099 <https://doi.org/10.7554/eLife.64099> | [Google Scholar](#)
- Barajas-Lopez C., Berezin I., Daniel E. E., Huizinga J. D (1989) **Pacemaker activity recorded in interstitial cells of Cajal of the gastrointestinal tract** *Am J Physiol* **257**:C830–835 <https://doi.org/10.1152/ajpcell.1989.257.4.C830> | [Google Scholar](#)
- Camborova P., Hubka P., Sulkova I., Hulin I (2003) **The pacemaker activity of interstitial cells of Cajal and gastric electrical activity** *Physiol Res* **52**:275–284 [PubMed](#) | [Google Scholar](#)
- Chevalier N. R (2018) **The first digestive movements in the embryo are mediated by mechanosensitive smooth muscle calcium waves** *Philos Trans R Soc Lond B Biol Sci* **373**:1759 <https://doi.org/10.1098/rstb.2017.0322> | [Google Scholar](#)
- Chevalier N. R., Ammouche Y., Gomis A., Teyssaire C., de Santa Barbara P., Faure S. (2020) **Shifting into high gear: how interstitial cells of Cajal change the motility pattern of the developing intestine** *Am J Physiol Gastrointest Liver Physiol* **319**:G519–g528 <https://doi.org/10.1152/ajpgi.00112.2020> | [Google Scholar](#)
- Chevalier N. R., Dacher N., Jacques C., Langlois L., Guedj C., Faklaris O (2019) **Embryogenesis of the peristaltic reflex** *J Physiol* **597**:2785–2801 <https://doi.org/10.1113/jp277746> | [Google Scholar](#)
- Cousins H. M., Edwards F. R., Hickey H., Hill C. E., Hirst G. D (2003) **Electrical coupling between the myenteric interstitial cells of Cajal and adjacent muscle layers in the guinea-pig gastric antrum** *J Physiol* **550**:829–844 <https://doi.org/10.1113/jphysiol.2003.042176> | [Google Scholar](#)
- Daniel E. E., Wang Y. F (1999) **Gap junctions in intestinal smooth muscle and interstitial cells of Cajal** *Microsc Res Tech* **47**:309–320 [https://doi.org/10.1002/\(SICI\)1097-0029\(19991201\)47:5<309::AID-JEMT2>3.0.CO;2-K](https://doi.org/10.1002/(SICI)1097-0029(19991201)47:5<309::AID-JEMT2>3.0.CO;2-K) | [Google Scholar](#)
- Duband J. L., Gimona M., Scatena M., Sartore S., Small J. V (1993) **Calponin and SM 22 as differentiation markers of smooth muscle: spatiotemporal distribution during avian embryonic development** *Differentiation* **55**:1–11 <https://doi.org/10.1111/j.1432-0436.1993.tb00027.x> | [Google Scholar](#)
- Fukui H., Chow R. W., Xie J., Foo Y. Y., Yap C. H., Minc N., Mochizuki N., Vermot J (2021) **Bioelectric signaling and the control of cardiac cell identity in response to mechanical forces** *Science* **374**:351–354 <https://doi.org/10.1126/science.abc6229> | [Google Scholar](#)
- Gabella G., Blundell D (1981) **Gap junctions of the muscles of the small and large intestine** *Cell Tissue Res* **219**:469–488 <https://doi.org/10.1007/BF00209987> | [Google Scholar](#)
- Gherghiceanu M., Popescu L. M (2005) **Interstitial Cajal-like cells (ICLC) in human resting mammary gland stroma. Transmission electron microscope (TEM) identification** *J Cell Mol*

Med **9**:893–910 <https://doi.org/10.1111/j.1582-4934.2005.tb00387.x> | [Google Scholar](#)

Graham H. K., Maina I., Goldstein A. M., Nagy N (2017) **Intestinal smooth muscle is required for patterning the enteric nervous system** *J Anat* **230**:567–574 <https://doi.org/10.1111/joa.12583> | [Google Scholar](#)

Grosse A. S., Pressprich M. F., Curley L. B., Hamilton K. L., Margolis B., Hildebrand J. D., Gumucio D. L (2011) **Cell dynamics in fetal intestinal epithelium: implications for intestinal growth and morphogenesis** *Development* **138**:4423–4432 <https://doi.org/10.1242/dev.065789> | [Google Scholar](#)

Huizinga J. D., Martz S., Gil V., Wang X. Y., Jimenez M., Parsons S (2011) **Two independent networks of interstitial cells of cajal work cooperatively with the enteric nervous system to create colonic motor patterns** *Front Neurosci* **5**:93 <https://doi.org/10.3389/fnins.2011.00093> | [Google Scholar](#)

Huizinga J. D., Thuneberg L., Kluppel M., Malysz J., Mikkelsen H. B., Bernstein A (1995) **W/kit gene required for interstitial cells of Cajal and for intestinal pacemaker activity** *Nature* **373**:347–349 <https://doi.org/10.1038/373347a0> | [Google Scholar](#)

Iino S., Horiguchi K (2006) **Interstitial cells of cajal are involved in neurotransmission in the gastrointestinal tract** *Acta Histochem Cytochem* **39**:145–153 <https://doi.org/10.1267/ahc.06023> | [Google Scholar](#)

Iino S., Horiguchi S., Horiguchi K., Nojyo Y (2007) **Interstitial cells of Cajal in the gastrointestinal musculature of W mutant mice** *Arch Histol Cytol* **70**:163–173 <https://doi.org/10.1679/aohc.70.163> | [Google Scholar](#)

Kito Y., Ward S. M., Sanders K. M (2005) **Pacemaker potentials generated by interstitial cells of Cajal in the murine intestine** *Am J Physiol Cell Physiol* **288**:C710–720 <https://doi.org/10.1152/ajpcell.00361.2004> | [Google Scholar](#)

Kluppel M., Huizinga J. D., Malysz J., Bernstein A (1998) **Developmental origin and Kit-dependent development of the interstitial cells of cajal in the mammalian small intestine** *Dev Dyn* **211**:60–71 [https://doi.org/10.1002/\(SICI\)1097-0177\(199801\)211:1<60::AID-AJA6>3.0.CO;2-5](https://doi.org/10.1002/(SICI)1097-0177(199801)211:1<60::AID-AJA6>3.0.CO;2-5) | [Google Scholar](#)

Komuro T., Tokui K., Zhou D. S (1996) **Identification of the interstitial cells of Cajal** *Histol Histopathol* **11**:769–786 [PubMed](#) | [Google Scholar](#)

Li Z. Q., Torgersen J., Ajami A., Mühleder S., Qin X. H., Husinsky W., Holnthoner W., Ovsianikov A., Stampfl J., Liska R. (2013) **Initiation efficiency and cytotoxicity of novel water-soluble two-photon photoinitiators for direct 3D microfabrication of hydrogels** *Rsc Advances* **3**:15939–15946 <https://doi.org/10.1039/c3ra42918k> | [Google Scholar](#)

Liu L. W., Thuneberg L., Huizinga J. D (1998) **Development of pacemaker activity and interstitial cells of Cajal in the neonatal mouse small intestine** *Dev Dyn* **213**:271–282 [https://doi.org/10.1002/\(SICI\)1097-0177\(199811\)213:3<271::AID-AJA4>3.0.CO;2-R](https://doi.org/10.1002/(SICI)1097-0177(199811)213:3<271::AID-AJA4>3.0.CO;2-R) | [Google Scholar](#)

Mei F., Han J., Huang Y., Jiang Z. Y., Xiong C. J., Zhou D. S (2009) **Plasticity of interstitial cells of cajal: a study in the small intestine of adult Guinea pigs** *Anat Rec (Hoboken)* **292**:985–993 <https://doi.org/10.1002/ar.20928> | [Google Scholar](#)

- Murai H., Tadokoro R., Sakai K., Takahashi Y (2015) **In ovo gene manipulation of melanocytes and their adjacent keratinocytes during skin pigmentation of chicken embryos** *Dev Growth Differ* **57**:232–241 <https://doi.org/10.1111/dgd.12201> | Google Scholar
- Rohr S., Kucera J. P., Kleber A. G (1998) **Slow conduction in cardiac tissue, I: effects of a reduction of excitability versus a reduction of electrical coupling on microconduction** *Circ Res* **83**:781–794 <https://doi.org/10.1161/01.res.83.8.781> | Google Scholar
- Rumessen J. J., Thuneberg L (1991) **Interstitial cells of Cajal in human small intestine. Ultrastructural identification and organization between the main smooth muscle layers** *Gastroenterology* **100**:1417–1431 [PubMed](#) | [Google Scholar](#)
- Rumessen J. J., Thuneberg L (1996) **Pacemaker cells in the gastrointestinal tract: interstitial cells of Cajal** *Scand J Gastroenterol Suppl* **216**:82–94 <https://doi.org/10.3109/00365529609094564> | [Google Scholar](#)
- Sanders K. M., Drumm B. T., Cobine C. A., Baker S. A (2024) **Ca(2+) dynamics in interstitial cells: foundational mechanisms for the motor patterns in the gastrointestinal tract** *Physiol Rev* **104**:329–398 <https://doi.org/10.1152/physrev.00036.2022> | [Google Scholar](#)
- Sanders K. M., Kito Y., Hwang S. J., Ward S. M (2016) **Regulation of Gastrointestinal Smooth Muscle Function by Interstitial Cells** *Physiology (Bethesda)* **31**:316–326 <https://doi.org/10.1152/physiol.00006.2016> | [Google Scholar](#)
- Sanders K. M., Publicover N. G., Ward S. M (1991) **Involvement of interstitial cells of Cajal in pacemaker activity of canine colon** *J Smooth Muscle Res* **27**:1–11 <https://doi.org/10.1540/jsmr.27.1> | [Google Scholar](#)
- Sanders K. M., Ward S. M., Koh S. D (2014) **Interstitial cells: regulators of smooth muscle function** *Physiol Rev* **94**:859–907 <https://doi.org/10.1152/physrev.00037.2013> | [Google Scholar](#)
- Schultz T., Daniel V., Daniel E. E (2003) **Does ICC pacing require functional gap junctions between ICC and smooth muscle in mouse intestine?** *Neurogastroenterol Motil* **15**:129–138 <https://doi.org/10.1046/j.1365-2982.2003.00401.x> | [Google Scholar](#)
- Shikaya Y., Takase Y., Tadokoro R., Nakamura R., Inaba M., Takahashi Y (2022) **Distribution Map of Peristaltic Waves in the Chicken Embryonic Gut Reveals Importance of Enteric Nervous System and Inter-Region Cross Talks Along the Gut Axis** *Front Cell Dev Biol* **10**:827079 <https://doi.org/10.3389/fcell.2022.827079> | [Google Scholar](#)
- Sun X., Yu B., Xu L., Dong W., Luo H (2006) **Interstitial cells of Cajal in the murine gallbladder** *Scand J Gastroenterol* **41**:1218–1226 <https://doi.org/10.1080/00365520600708800> | [Google Scholar](#)
- Takaki M (2003) **Gut pacemaker cells: the interstitial cells of Cajal (ICC)** *J Smooth Muscle Res* **39**:137–161 <https://doi.org/10.1540/jsmr.39.137> | [Google Scholar](#)
- Takayama I., Horiguchi K., Daigo Y., Mine T., Fujino M. A., Ohno S (2002) **The interstitial cells of Cajal and a gastroenteric pacemaker system** *Arch Histol Cytol* **65**:1–26 <https://doi.org/10.1679/aohc.65.1> | [Google Scholar](#)
- Takeda Y., Ward S. M., Sanders K. M., Koh S. D (2005) **Effects of the gap junction blocker glycyrrhetic acid on gastrointestinal smooth muscle cells** *Am J Physiol Gastrointest Liver Physiol* **288**:G832–841 <https://doi.org/10.1152/ajpgi.00389.2004> | [Google Scholar](#)

Thomsen L., Robinson T. L., Lee J. C., Farraway L. A., Hughes M. J., Andrews D. W., Huizinga J. D (1998) **Interstitial cells of Cajal generate a rhythmic pacemaker current** *Nat Med* **4**:848–851 <https://doi.org/10.1038/nm0798-848> | [Google Scholar](#)

Torihashi S., Fujimoto T., Trost C., Nakayama S (2002) **Calcium oscillation linked to pacemaking of interstitial cells of Cajal: requirement of calcium influx and localization of TRP4 in caveolae** *J Biol Chem* **277**:19191–19197 <https://doi.org/10.1074/jbc.M201728200> | [Google Scholar](#)

Torihashi S., Nishi K., Tokutomi Y., Nishi T., Ward S., Sanders K. M (1999) **Blockade of kit signaling induces transdifferentiation of interstitial cells of cajal to a smooth muscle phenotype** *Gastroenterology* **117**:140–148 [https://doi.org/10.1016/s0016-5085\(99\)70560-3](https://doi.org/10.1016/s0016-5085(99)70560-3) | [Google Scholar](#)

Van Hoorick J., Gruber P., Markovic M., Rollot M., Graulus G. J., Vagenende M., Tromayer M., Van Erps J., Thienpont H., Martins J. C., Baudis S., Ovsianikov A., Dubruel P., Van Vlierberghe S. (2018) **Highly Reactive Thiol-Norbornene Photo-Click Hydrogels: Toward Improved Processability** *Macromol Rapid Commun* **39**:e1800181 <https://doi.org/10.1002/marc.201800181> | [Google Scholar](#)

Wang Q., Wang K., Solorzano-Vargas R. S., Lin P. Y., Walther C. M., Thomas A. L., Martín M. G., Dunn J. C. Y (2018) **Bioengineered intestinal muscularis complexes with long-term spontaneous and periodic contractions** *PLoS One* **13**:e0195315 <https://doi.org/10.1371/journal.pone.0195315> | [Google Scholar](#)

Yagasaki R., Shikaya Y., Kawachi T., Inaba M., Takase Y., Takahashi Y (2022) **Newly raised anti-c-Kit antibody visualizes morphology of interstitial cells of Cajal in the developing gut of chicken embryos** *Dev Growth Differ* **64**:446–454 <https://doi.org/10.1111/dgd.12808> | [Google Scholar](#)

Yang P., Yu Z., Gandahi J. A., Bian X., Wu L., Liu Y., Zhang L., Zhang Q., Chen Q (2012) **The identification of c-Kit-positive cells in the intestine of chicken** *Poult Sci* **91**:2264–2269 <https://doi.org/10.3382/ps.2011-02076> | [Google Scholar](#)

Zeng X., Komanome Y., Kawasaki T., Inada K., Jonaitis J., Pulver S. R., Kazama H., Nose A (2021) **An electrically coupled pioneer circuit enables motor development via proprioceptive feedback in Drosophila embryos** *Curr Biol* **31**:5327–5340 <https://doi.org/10.1016/j.cub.2021.10.005> | [Google Scholar](#)

Author information

Rei Yagasaki

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan, WPI Nano Life Science Institute (WPI-NanoLSI), Kanazawa University, Kanazawa, Japan

Ryo Nakamura

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan, Ushimado Marine Institute, Okayama University, Okayama, Japan
ORCID iD: [0000-0002-0761-0704](https://orcid.org/0000-0002-0761-0704)

Yuuki Shikaya

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan, Graduate School of Science, University of Hyogo, Hyogo, Japan
ORCID iD: [0009-0009-3544-9725](https://orcid.org/0009-0009-3544-9725)

Ryosuke Tadokoro

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan, Department of Bioscience, Okayama University of Science, Okayama, Japan
ORCID iD: [0000-0001-7556-4263](https://orcid.org/0000-0001-7556-4263)

Ruolin Hao

Institute for Life and Medical Sciences, Kyoto University, Kyoto, Japan, Department of Polymer Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Japan

Zhe Wang

Institute for Life and Medical Sciences, Kyoto University, Kyoto, Japan, Department of Polymer Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Japan
ORCID iD: [0000-0002-6717-0253](https://orcid.org/0000-0002-6717-0253)

Mototsugu Eiraku

Institute for Life and Medical Sciences, Kyoto University, Kyoto, Japan, Department of Polymer Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Japan
ORCID iD: [0000-0002-9433-9085](https://orcid.org/0000-0002-9433-9085)

Masafumi Inaba

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan
ORCID iD: [0000-0001-6896-5819](https://orcid.org/0000-0001-6896-5819)

Yoshiko Takahashi

Department of Zoology, Graduate School of Science, Kyoto University, Kyoto, Japan
ORCID iD: [0000-0002-1596-7527](https://orcid.org/0000-0002-1596-7527)

For correspondence: yotayota@develop.zool.kyoto-u.ac.jp

Editors

Reviewing Editor

Akinao Nose

The University of Tokyo, Chiba-ken, Japan

Senior Editor

Didier Stainier

Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

Reviewer #1 (Public review):

Summary:

In this study, the authors developed an organoid system containing smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs; pacemaker cells), but few enteric neurons. This system generates rhythmic contractions similar to those observed in the developing gut. The stereotypical arrangement of SMCs and ICCs within the organoid allowed the authors to

identify these cell types without the need for antibody staining. Leveraging this feature, they used calcium imaging and pharmacological approaches to investigate how calcium transients develop through interactions between the two cell types.

The authors first show that calcium transients are synchronized among ICC-ICC, SMC-SMC, and SMC-ICC pairs. They then used gap junction inhibitors to suggest that gap junctions are specifically involved in ICC-to-SMC signaling. Finally, they applied inhibitors of myosin II and L-type Ca^{2+} channels to demonstrate that SMC contraction is crucial for the generation of rhythmic activity in ICCs, suggesting the presence of SMC-to-ICC signaling. Additionally, they show that two organoids become synchronized upon fusion, with SMCs mediating this synchronization.

Strengths:

The organoid system provides a useful model for studying the specific roles of SMCs and ICCs in live samples.

Weaknesses:

Since all functional analyses were conducted pharmacologically *in vitro*, the findings need to be further validated through genetic approaches *in vivo* in future studies.

<https://doi.org/10.7554/eLife.97860.2.sa3>

Reviewer #2 (Public review):

Summary:

In this study, Yagasaki et al. describe an organoid system to study the interactions between smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs). While these interactions are essential for the control of rhythmic intestinal contractility (i.e., peristalsis), they are poorly understood, largely due to the complexity of and access to the *in vivo* environment and the inability to co-culture these cell types *in vitro* for long term under physiological conditions. The "gut contractile organoids" organoids described herein are reconstituted from stromal cells of the fetal chicken hindgut that rapidly reorganize into multilayered spheroids containing an outer layer of smooth muscle cells and an inner core of interstitial cells. The authors demonstrate that they contract cyclically and additionally use calcium imaging to show that these contractions occur concomitantly with calcium transients that initiate in the interstitial cell core and are synchronized within the organoid and between ICCs and SMCs. Furthermore, they use several pharmacological inhibitors to show that these contractions are dependent upon non-muscle myosin activity and, surprisingly, independent of gap junction activity. Finally, they develop a 3D hydrogel for the culturing of multiple organoids and found that they synchronize their contractile activities through interconnecting smooth muscle cells, suggesting that this model can be used to study the emergence of pacemaking activities. Overall, this study provides a relatively easy-to-establish organoid system that will be of use in studies examining the emergence of rhythmic peristaltic smooth muscle contractions and how these are regulated by interstitial cell interactions. However, further validation and quantification will be necessary to conclusively determine show the cellular composition of the organoids and how reproducible their behaviors are.

Strengths:

This work establishes a new self-organizing organoid system that can easily be generated from the muscle layers of the chick fetal hindgut to study the emergence of spontaneous smooth muscle cell contractility. A key strength of this approach is that the organoids seem to contain few cell types (though more validation is needed), namely smooth muscle cells (SMCs)

and interstitial cells of Cajal (ICCs). These organoids are amenable to live imaging of calcium dynamics as well as pharmacological perturbations for functional assays, and since they are derived from developing tissues, the emergence of the interactions between cell types can be functionally studied. Thus, the gut contractile organoids represent a reductionist system to study the interactions between SMCs and ICCs in comparison to the more complex in vivo environment, which has made studying these interactions challenging.

Weaknesses:

The study lacks complementary in vivo experiments, but these will be exciting to follow up in future studies.

<https://doi.org/10.7554/eLife.97860.2.sa2>

Reviewer #3 (Public review):

Summary:

The paper presents a novel contractile gut organoid system that allows for in vitro studying of rudimentary peristaltic motions in embryonic tissues by facilitating GCaMP-live imaging of Ca²⁺ dynamics, while highlighting the importance and sufficiency of ICC and SMC interactions in generating consistent contractions reminiscent of peristalsis. It also argues that ENS at later embryonic stages might not be necessary for coordination of peristalsis.

Strengths:

The manuscript by Yagasaki, Takahashi, and colleagues represents an exciting new addition to the toolkit available for studying fundamental questions in the development and physiology of the hindgut. The authors carefully lay out the protocol for generating contractile gut organoids from chick embryonic hindgut and perform a series of experiments that illustrate the broader utility of these organoids for studying the gut. This reviewer is highly supportive of the manuscript following highly responsive revisions in response to prior reviewer feedback.

<https://doi.org/10.7554/eLife.97860.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This valuable study reports the development of a novel organoid system for studying the emergence of autorhythmic gut peristaltic contractions through the interaction between interstitial cells of Cajal and smooth muscle cells. While the utility of the organoids for studying hindgut development is well illustrated by showing, for example, a previously unappreciated potential role for smooth muscle cells in regulating the firing rate of interstitial cells of Cajal, some of the functional analyses are incomplete. There are some concerns about the specificity and penetrance of perturbations and the reproducibility of the phenotypes. With these concerns properly addressed, this paper will be of interest to those studying the development and physiology of the gut.

We greatly appreciate constructive comments raised by the Editors and all the Reviewers. We have newly conducted pharmacological experiments using Nifedipine, a L-type Ca²⁺ blocker known to operate in smooth muscles (new Fig 7). The treatment abrogated not only the

oscillation of SMCs but also that in ICCs, further corroborating our model that not only ICC-to-SMC interactions but also the reverse direction, namely SMC-to-ICC feedback signals, are operating to achieve coordinated/stable rhythm of gut contractile organoids.

Concerning the issues of the specificity and penetrance in pharmacological experiments with gap junction inhibitors, we have carefully re-examined effects by multiple blockers (CBX and 18b-GA) at different concentrations (new Fig 5D and Fig. S3B). We have newly found that: (1) the effects observed by CBX (100 μ M) that the latency of Ca^{2+} peaks between ICCs (preceding) and SMCs (following) was abolished are not seen by 18b-GA at any concentrations including 100 μ M, implying that the latency of Ca^{2+} peaks between these cells is governed by connexin(s) that are not inhibited by 18bGA. Such difference in inhibiting effects by these two drugs were previously reported in multiple model systems including guts (Daniel et al., 2007; Parsons & Huizinga, 2015; Schultz et al., 2003).

Regarding the penetrance of the drugs, we have carried out earlier administration (Day 3) of the gap junction inhibitor, either CBX (100 μ M) or 18b-GA (100 μ M), in the course of organoidal formation in culture when cells are still at 2D to exclude a possible penetrance problem (new Fig. S3C). These treatments render no or little effects to the patterns of organoidal contractions in a way similar to the drug administration at Day 7. As already shown in the first version, CBX (100 μ M) eliminates the latency of Ca^{2+} peaks, we believe that this drug successfully penetrates into the organoid and exerts its specific effects.

Unfortunately, due to very unstable condition in climate including extreme heat and sporadically occurring bird flu epidemic since the last summer in Japan, the poultry farm must have faced problems. In the course of revision experiments, we got in a serious trouble at multiple times with unhealthy eggs/embryos lasting from last summer until present. These unfortunate incidents did not allow us to engage in the revision experiments as fully as we originally planned. Nevertheless, we did our very best within a limited time frame, and we believe that the revised version is suitable as a final version of an eLife article.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

In this study, the authors developed an organoid system that contains smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs; pacemaker) but few enteric neurons, and generates rhythmic contractions as seen in the developing gut. The stereotypical arrangements of SMCs and ICCs in the organoid allowed the authors to identify these cell types in the organoid without antibody staining. The authors took advantage of this and used calcium imaging and pharmacology to study how calcium transients develop in this system through the interaction between the two types of cells. The authors first show that calcium transients are synchronized between ICC-ICC, SMC-SMC, and SMC-ICC. They then used gap junction inhibitors to suggest that gap junctions are specifically involved in ICC-to-SMC signaling. Finally, the authors used an inhibitor of myosin II to suggest that feedback from SMC contraction is crucial for the generation of rhythmic activities in ICCs. The authors also show that two organoids become synchronized as they fuse and SMCs mediate this synchronization.

Strengths:

The organoid system offers a useful model in which one can study the specific roles of SMCs and ICCs in live samples.

Thank you very much for the constructive comments.

Weaknesses:

Since only one blocker each for gap junction and myosin II was used, the specificities of the effects were unclear.

We appreciate these comments. We have addressed those of “weaknesses” as described in “Responses to the eLife assessment” (please see above).

Reviewer #2 (Public Review):

Summary:

In this study, Yagasaki et al. describe an organoid system to study the interactions between smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs). While these interactions are essential for the control of rhythmic intestinal contractility (i.e., peristalsis), they are poorly understood, largely due to the complexity of and access to the in vivo environment and the inability to co-culture these cell types in vitro for long term under physiological conditions. The "gut contractile organoids" organoids described herein are reconstituted from stromal cells of the fetal chicken hindgut that rapidly reorganize into multilayered spheroids containing an outer layer of smooth muscle cells and an inner core of interstitial cells. The authors demonstrate that they contract cyclically and additionally use calcium imaging to show that these contractions occur concomitantly with calcium transients that initiate in the interstitial cell core and are synchronized within the organoid and between ICCs and SMCs. Furthermore, they use several pharmacological inhibitors to show that these contractions are dependent upon non-muscle myosin activity and, surprisingly, independent of gap junction activity. Finally, they develop a 3D hydrogel for the culturing of multiple organoids and found that they synchronize their contractile activities through interconnecting smooth muscle cells, suggesting that this model can be used to study the emergence of pacemaking activities. Overall, this study provides a relatively easy-to-establish organoid system that will be of use in studies examining the emergence of rhythmic peristaltic smooth muscle contractions and how these are regulated by interstitial cell interactions. However, further validation and quantification will be necessary to conclusively determine show the cellular composition of the organoids and how reproducible their behaviors are.

Strengths:

This work establishes a new self-organizing organoid system that can easily be generated from the muscle layers of the chick fetal hindgut to study the emergence of spontaneous smooth muscle cell contractility. A key strength of this approach is that the organoids seem to contain few cell types (though more validation is needed), namely smooth muscle cells (SMCs) and interstitial cells of Cajal (ICCs). These organoids are amenable to live imaging of calcium dynamics as well as pharmacological perturbations for functional assays, and since they are derived from developing tissues, the emergence of the interactions between cell types can be functionally studied. Thus, the gut contractile organoids represent a reductionist system to study the interactions between SMCs and ICCs in comparison to the more complex in vivo environment, which has made studying these interactions challenging.

Thank you very much for the constructive comments.

Weaknesses:

The study falls short in the sense that it does not provide a rigorous amount of evidence to validate that the gut organoids are made of bona fide smooth muscle cells and ICCs. For example, only two "marker" proteins are used to support the claims of cell identity of SMCs and ICCs. At the same time, certain aspects of the data are not quantified sufficiently to appreciate the variance of organoid rhythmic contractility. For example, most contractility plots show the trace for a single organoid. This leads to a concern for how reproducible certain aspects of the organoid system (e.g. wavelength between contractions/rhythm) might be, or how these evolve uniquely over time in culture. Furthermore, while this study might be able to capture the emergence of ICC-SMC interactions as they related to muscle contraction and pacemaking, it is unclear how these interactions relate to adult gastrointestinal physiology given that the organoids are derived from fetal cells that might not be fully differentiated or might have distinct functions from the adult. Finally, despite the strength of this system, discoveries made in it will need to be validated in vivo. Thank you very much for the comments, which are helpful to improve our MS. In the revised version, we have additionally used antibody against desmin, known to be a maker for mature SMCs (new Fig 3B). The signal is seen only in the peripheral cells overlapping with the αSMA staining (line 169-170).

Concerning the reproducibility, while contractility changes were shown for a representative organoid in the original version, experiments had been carried out multiple times, and consistent data were reproduced as already mentioned in the text of the first version of MS. However, we agree with this reviewer that it must be more convincing if we assess quantitatively. We have therefore conducted quantitative assessments of organoidal contractions and Ca^{2+} transients (new Fig. 2B, new Fig. 4D, new Fig 5D, E, new Fig. 6B, new Fig. 7B, new Fig. 8C, new Fig. S2, S3). Details such as repeats of experiments and size of specimens are carefully described in the revised version (Figure legends)

In particular, in place of contraction numbers/time, we have plotted "contraction intervals" between two successive peaks (Fig. 2B and others). Actually, with your suggestion, we have tried to perform a periodicity analysis of organoid contractions. Unfortunately, no clear value has been obtained, probably because the contractions/ Ca^{2+} transitions are not as "regularly periodical" as seen in conventional physics. This led us to perform the peak-interval analysis. Methods to quantify the contraction intervals are carefully explained in the revised version.

As already mentioned in the "Our provisional responses" following the receipt of Reviewers' comments, we agree that our organoids derived from embryonic hind gut (E15) might not necessarily recapitulate the full function of cells in adult. However, it has well been accepted in the field of developmental biology that studies with embryonic tissue/cells make a huge contribution to unveil complicated physiological cell functions. Nevertheless, we have carefully considered in the revised version so that the MS would not send misleading messages. We agree that in vivo validation of our gut contractile organoid must be wonderful, and this is a next step to go.

Reviewer #3 (Public Review):

Summary:

The paper presents a novel contractile gut organoid system that allows for in vitro studying of rudimentary peristaltic motions in embryonic tissues by facilitating GCaMP6s imaging of Ca^{2+} dynamics, while highlighting the importance and sufficiency of ICC and SMC interactions in generating consistent contractions reminiscent of peristalsis. It also argues that ENS at later embryonic stages might not be necessary for coordination of peristalsis.

Strengths:

The manuscript by Yagasaki, Takahashi, and colleagues represents an exciting new addition to the toolkit available for studying fundamental questions in the development and physiology of the hindgut. The authors carefully lay out the protocol for generating contractile gut organoids from chick embryonic hindgut, and perform a series of experiments that illustrate the broader utility of these organoids for studying the gut. This reviewer is highly supportive of the manuscript, with only minor requests to improve confidence in the findings and broader impact of the work. These are detailed below.

Thank you very much for the constructive comments.

Weaknesses:

(1) Given that the literature is conflicting on the role GAP junctions in potentiating communication between intestinal cells of Cajal (ICCs) and smooth muscle cells (SMCs), the experiments involving CBX and 18Beta-GA are well-justified. However, because neither treatment altered contractile frequency or synchronization of Ca⁺⁺ transients, it would be important to demonstrate that the treatments did indeed inhibit GAP junction function as administered. This would strengthen the conclusion that GAP junctions are not required, and eliminate the alternative explanation that the treatments themselves failed to block GAP junction activity.

Thank you for these comments, and we agree. In the revised version, we have verified the drugs, CBX and 18b-GA, using dissociated embryonic heart cells in culture, a well-established model for the gap junction study (new Fig. S3D, line 237-239). Expectedly, both inhibitors abrogate the rhythmic beats of heart cells, and importantly, cells' beats resume after wash-out of the drug.

(2) Given that 5uM blebbistatin increases the frequency of contractions but 10uM completely abolishes contractions, confirming that cell viability is not compromised at the higher concentration would build confidence that the phenotype results from inhibition of myosin activity. One could either assay for cell death, or perform washout experiments to test for recovery of cyclic contractions upon removal of blebbistatin. The latter may provide access to other interesting questions as well. For example, do organoids retain memory of their prior setpoint or arrive at a new firing frequency after washout?

We greatly appreciate these suggestions and also interesting ideas to explore! In the revised version, we have newly conducted washout experiments (new Fig. 6B) (10 μ M drug is washed-out from culture medium), and found that contractions resume, showing that cell viability is not compromised at 10 μ M concentration (line 257-259). Intriguingly, the resumed rhythm appears more regular than that before drug administration. Thus, the contraction rhythm of the organoid might be determined by cell-cell interactions at any given time rather than by memory of their prior setpoint. This is an interesting issue we would like to further explore in the future. These issues, although potentially interesting, are not mentioned in the text of the revised version, since it is too early to interpret these observations.

(3) Regulation of contractile activity was attributed to ICCs, with authors reasoning that Tuj1+ enteric neurons were only present in organoids in very small numbers (~1%).

However, neuronal function is not strictly dependent on abundance, and some experimental support for the relative importance of ICCs over Tuj1+ cells would strengthen a central assumption of the work that ICCs are the predominant cell type regulating organoid contraction. For example, one could envision forming organoids from embryos in which neural crest cells have been ablated via microdissection or

targeted electroporation. Another approach would be ablation of Tuj1+ cells from the formed organoids via tetrodotoxin treatment. The ability of organoids to maintain rhythmic contractile activity in the total absence of Tuj1+ cells would add confidence that the ICCs are indeed the driver of contractility in these organoids.

We agree. In the revised version, we have conducted TTX administration (new Fig. S2C). Changes in contractility by this treatment is not detected, supporting the argument that neural cells/activities are not essential for rhythmic contractions of the organoid (line 178-181).

(4) Given the implications of a time lag between Ca⁺⁺ peaks in ICCs and SMCs, it would be important to quantify this, including standard deviations, rather than showing representative plots from a single sample.

In the revised version, we have elaborated a series of quantitative assessments as mentioned above (please see our responses to the “eLife assessments” at the beginning of these correspondences). The latency between Ca²⁺ peaks in ICCs and SMCs is shown in new Fig. 4D, in which measured value is 700 msec-terraced since the time-lapse imaging was performed with 700 msec intervals (as already described in the first version).

117 peaks for 14 organoids have been assessed (line 218).

(5) To validate the organoid as a faithful recreation of in vivo conditions, it would be helpful for authors to test some of the more exciting findings on explanted hindgut tissue. One could explant hindguts and test whether blebbistatin treatment silences peristaltic contractions as it does in organoids, or following RCAS-GCAMP infection at earlier stages, one could test the effects of GAP junction inhibitors on Ca⁺⁺ transients in explanted hindguts. These would potentially serve as useful validation for the gut contractile organoid, and further emphasize the utility of studying these simplified systems for understanding more complex phenomena in vivo.

Thank you very much for insightful comments. We would love to explore these issues in near future. Just a note is that it was previously reported that Nifedipine silences peristaltic contractions in ex-vivo cultured gut (Chevalier et al., 2024; Der et al., 2000).

(6) Organoid fusion experiments are very interesting. It appears that immediately after fusion, the contraction frequency is markedly reduced. Authors should comment on this, and how it changes over time following fusion. Further, is there a relationship between aggregate size and contractile frequency? There are many interesting points that could be discussed here, even if experimental investigation of these points is left to future work.

It would indeed be interesting to explore how cell communications affect/determine the contraction rhythm, and our novel organoids must serve as an excellent model to address these fundamental questions. We have observed multiple times that when two organoids fuse, they undergo “pause”, and resume coordinated contractions as a whole, and we have mentioned such notice briefly in the revised version (line 282). To know what is going on during this pause time should be tempting. In addition, we have an impression that the larger in size organoids grow, the slower rhythm they count. We would love to explore this in near future.

(7) Minor: As seen in Movie 6 and Figure 6A, 5uM blebbistatin causes a remarkable increase in the frequency of contractions. Given the regular periodicity of these contractions, it is a surprising and potentially interesting finding, but authors do not comment on it. It would be helpful to note this disparity between 5 and 10 uM

treatments, if not to speculate on what it means, even if it is beyond the scope of the present study to understand this further.

We assume that the increase in the frequency of contractions at 5 μ M might be due to a shorter refractory period caused by a decreasing magnitude (amplitude) of contraction. We have made a short description in the revised text (line 256-257).

(8) Minor: While ENS cells are limited in the organoid, it would be helpful to quantify the number of SMCs for comparison in Supplemental Figure S2. In several images, the number of SMCs appears quite limited as well, and the comparison would lend context and a point of reference for the data presented in Figure S2B.

In the revised version, the number of SMCs has been counted and added in Fig. S2B. Contrary to that SMCs are more abundant than ICCs in an intact gut, the proportion is reversed in our organoid (line 181-183). It might due to treatments during cell dissociation/plating.

(9) Minor: additional details in the Figure 8 legend would improve interpretation of these results. For example, what is indicated in orange signal present in panels C, G and H? Is this GCaMP?

We apologize for this confusion. In the revised version, we have added labeling directly in the photos of new Fig. 9 (old Fig. 8). For C, G and H, the left photo is mRuby3+GCaMP6s, and the right one is GCaMP6s only.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

I have a few comments for the authors to consider:

(1) Figure 4C: The authors propose that calcium signals propagate from ICC to SMC based on the results presented in this figure. While it is observed that the peak of the calcium signal in ICC precedes that in SMC, it's worth noting that the onset of the rise in calcium signals occurs simultaneously in ICC and SMC. Doesn't this suggest that they are activated simultaneously? The latency observed for the peaks of calcium signals could reflect different kinetics of the rise in calcium concentration in the two types of cells rather than the order of calcium signal propagation.

We greatly appreciate these comments. We have re-examined kinetics of GCaMP signals in ICC and SMC, but we did not succeed in validating rise points precisely. We agree that the possibility that the rise in calcium signals could be occurring simultaneously. To clarify these issues, analyses with higher resolution is required, such as using GCaMP6f or GCaMP7/8. Nevertheless, the disappearance of the latency of Ca^{2+} peak by CBX implies a role of gap junction in ICC to SMC signaling. In the revised version, we replaced the wording “rise” by “peak” when the latency is discussed.

(2) Figure 5C: The specific elimination of the latency in the calcium signal peaks between ICC and SMC is interesting. However, I am curious about how gap junction inhibitors specifically eliminate the latency between ICC and SMC without affecting other aspects of calcium transients in these cells, such as amplitude and synchronization among ICCs and/or SMCs. Readers of the manuscript would expect some discussion on possible mechanisms underlying this specificity. Additionally, I wonder if the elimination of the latency was observed consistently across all samples examined. The authors should provide information on the frequency and number of samples examined, and whether the elimination occurs when 18-beta-GA is used.

In the revised version, we have elaborated quantitative demonstration. For the effects by CBX on latency or Ca^{2+} peaks, a new graph has been added to new Fig 5, in which 100 μM eliminated the latency. Intriguingly, the latency appears to be attributed to a gap junction that is not inhibited by 18-beta-GA (please see new Fig. S3E). As already mentioned above, inhibiting activity of both CBX and 18-beta-GA has been verified using dissociated cells of embryonic heart, a popular model for gap junction studies.

At present, we do not know how gap junction(s) contribute to the latency of Ca^{2+} peaks without affecting synchronization among ICCs and/or SMCs (we have not addressed amplitude of the oscillation in this study). Actually, it was surprising to us to find that GJ's contribution is very limited. We do not exclude the importance of GJs, and currently speculate that GJs might be important for the initiation of contraction/oscillation signals, whereas the requirement of GJs diminishes once the ICC-SMC interacting rhythm is established. What we observed in this study might be the synchronization signals AFTER these interactions are established (Day 7 of organoidal culture). Upon the establishment, it is possible that mechanical signaling elicited by smooth muscles' contraction might become prominent as a mediator for the (stable) synchronization, as implicated by experiments with blebbistatin and Nifedipin, the latter being newly added to the revised version (new Fig. 7). We have added such speculation, although briefly in Discussion (line 374-377)

(3) Figure 6: The significant effects of blebbistatin on calcium dynamics in both ICC and SMC are intriguing. However, since only one blocker is utilized, the specificity of the effects is unclear. If other blockers for muscle contraction are available, they should be employed. Considering that a rise in calcium concentration precedes contraction, calcium transients should persist even if muscle contraction is inhibited. One concern is whether blebbistatin inadvertently rendered the cells unhealthy. The authors should demonstrate at least that contraction and calcium transients recover after removal of the drug. The frequency and number of samples examined should be shown, as requested for Figure 5C above.

Thank you for these critical comments. A possible harmfulness of the drugs was also raised by other reviewers, and we have therefore conducted wash-out experiments in the revised version (new Fig. 6B). Contractions resume after wash-out showing that cell viability is not compromised at 10 μM concentration. The number of samples examined has been described more explicitly in the revised version. Regarding the blocker of SMC, we have newly carried out pharmacological assays using nifedipine, a blocker of a L-type Ca^{2+} channel known to operate in smooth muscle cells (new Fig 7) (Chevalier et al., 2024; Der et al., 2000). As already explained in the "Responses to eLife assessment", the treatment abrogated ICCs' rhythm and synchronous Ca^{2+} transients between ICCs and SMCs, further corroborating our model that not only ICC-to-SMC interactions but also SMC-to-ICC feedback signals are operating to achieve coordinated/stable rhythm of gut contractile organoids of Day 7 culture (please also see our responses shown above for Comment (2)).

Reviewer #2 (Recommendations For The Authors):

Major:

(1) The claim that organoids contain functional SMCs and ICCs is insufficient as it currently relies on only c-Kit and αSMA antibodies. This conclusion could be additionally supported by staining with other markers of contractile smooth muscle (e.g. TAGLN and MYH14) and an additional accepted marker of ICCs (e.g. ANO1/TMEM16). Moreover, it should be demonstrated whether these cells are PDGFRA+, as PDGFRA is a known marker of other mesenchymal fibroblast cell types. These experiments would additionally rule out whether these cells were simply less differentiated myofibroblasts. Given that there

might not be available antibodies that react with chicken protein versions, the authors could support their conclusions using alternative approaches, such as fluorescent in situ hybridization. A more thorough approach, such as single-cell RNA sequencing to compare the cell composition of the in vitro organoids to the in vivo colon, would fully justify the use of these organoids as a system for studying in vivo cell physiology.

With these suggestions provided, we have newly stained contractile organoids with anti-desmin antibody, known to be a marker for differentiated SMCs. As shown in new Fig. 3B, desmin-positive cells perfectly overlapped with aSMA-staining, indicating that the peripherally enclosing cells are SMCs. Regarding the interior cells, as this Reviewer concerned, there are no antibodies against ANO1/TMEM16 which are available for avian specimens. The anti- c-Kit antibody used in this study is what we raised in our hands by spending years (Yagasaki et al., 2021)), in which the antibody was carefully validated in intact guts of chicken embryos by multiple methods including Western Blot analyses, immunostaining, and in situ hybridization. We have attempted several times to perform organoidal whole-mount in situ hybridization for expression of PDGFR α , but we have not succeeded so far. In addition, as explained to the Editor, the very unhealthy condition of purchased eggs these past 7 months did not allow us to continue any further. We are planning to interrogate cell types residing in the central area of the organoid, results of which will be reported in a separate paper in near future.

(2) The key ICC-SMC relationship and physiological interaction seems to arise developmentally, but the mechanisms of this transition are not well defined (Chevalier 2020). To further support the claim that ICC-SMC interactions can be interrogated in this system, this study would benefit from establishing organoids at distinct developmental stages to (a) show that they have unique contractile profiles, and (b) demonstrate that they evolve over time in vitro toward an ICC-driven mechanism.

We agree with these comments. We tried to prepare gut contractile organoids derived from different stages of development, and we had an impression that slightly younger hindguts are available for the organoid preparations. In addition, not only the hindgut, but also midgut and caecum also yield organoids. However, since formed organoids derived from these “non-E15 hindgut” vary substantially in shapes, contraction frequencies/amplitudes etc., we are currently not ready to report these preliminary observations. Instead, we decided to optimize and elaborate in vitro culture conditions by focusing on the E15 hindgut, which turned out to be most stable in our hands. Nevertheless, it is tempting to see how organoid evolves over time during gut development.

(3) This manuscript would be greatly enhanced by a functional examination of the prospective organoid ICCs. For example, the authors could test whether the c-Kit inhibitor Imatinib, which has previously been used to impair ICC differentiation and function in the developing chick gut (Chevalier 2020), has an effect on contractility at different stages.

Following the paper of (Chevalier 2020), we had already conducted similar experiments with Imatinib in the culture with our organoids, but we did not see detectable effects. In that paper, the midgut of younger embryos was used, whereas we used E15 hindgut to prepare organoids. It would be interesting to see if we add Imatinib earlier during organoidal formation, and this is a next step to go.

(4) It is claimed that there is a 690s msec delay in SMC spike relative to ICC spike, however, it is unclear where this average is derived from and whether the organoid calcium trace shown in Figure 4C is representative of the data. The latency quantification

should be shown across multiple organoids, and again in the case of carbenoxolone treatment, to better understand the variations in treatment.

We apologize that the first version failed to clearly demonstrate quantitative assessments. In the revised version, we have elaborated quantitative assessments (117 peaks for 14 organoids) (line 216-218). In new Fig. 4D, measured value is 700 msec terraced since as already mentioned in the first version, the time-lapse imaging was performed with 700 msec intervals.

(5) As above, a larger issue is that only single traces are shown for each organoid. This makes it challenging to understand the variance in contractile properties across multiple organoids. While contraction frequencies are shown several times, the manuscript would benefit from additional quantifications, such as rhythm (average wavelength between events) in control and perturbed conditions.

We have substantially elaborated quantitative assessments (please also see our responses to the “Public Review”). In particular, in place of contraction numbers/time, we have plotted “contraction intervals” between two successive peaks (Fig. 2B and others). Actually, we have tried to perform a periodicity analysis of organoid contractions. Unfortunately, no clear value has been obtained, probably because the contractions/ Ca^{2+} transitions are not as “regularly periodical” as seen in conventional physics. This led us to perform the peak-interval analysis. Methods to quantify the contraction intervals are carefully explained in the revised version.

(6) The synchronicity observed between ICCs and SMCs within the organoid is interesting, and should be emphasized by making analyses more quantitative so as to understand how consistent and reproducible this phenomenon is across organoids. Moreover, one of the most exciting parts of the study is the synchronicity established between organoids in the hydrogel system, but it is insufficiently quantified. For example, how rapidly is pacemaking synchronization achieved?

As we replied above to (5), and described in the responses to the “Public Review”, we have substantially elaborated quantitative assessments in the revised version. Concerning the synchronicity between ICCs and SMCs, our data explicitly show that as long as the organoid undergoes healthy contraction, they perfectly match their rhythm (Fig. 4) making it difficult to display quantitatively. Instead, to demonstrate such synchronicity more convincingly, we have carefully described the number of peaks and the number of independent organoids we analyzed in each of Figure legends. In the experiments with hydrogels, the time required for two organoids to start/resume synchronous contraction varies greatly. For example, for the experiment shown in new Fig 9F, it takes 1 day to 2 days for cells crawling out of organoids and cover the surface of the hydrogel. In the experiments shown in new Fig. 8, two organoids undergo “pause” before resuming contractions. In the revised version, we have briefly mentioned our notice and speculation that active cell communications take place during this pausing time, (line 282-283 in Result and line 437-439 in Discussion). We agree with this reviewer saying that the pausing time is potentially very interesting. However, it is currently difficult to quantify these phenomena. More elaborate experimental design might be needed.

(7) Smooth muscle layers in vivo are well organized into circular and longitudinal layers. To establish physiological relevance, the authors should demonstrate if these organoids have multiple layers (though it looks like just a single outer layer) and if they show supracellular organization across the organoid.

The immunostaining data suggest that peripherally lining cells are of a single layer, and we assume that they might be aligned in register with contracting direction. However, to clarify these issues, observation with higher resolution would be required.

(8) To further examine whether the organoids contain true functional ICCs, the authors should test whether their calcium transients are impacted by inhibitors of L-type calcium channels, such as nifedipine and nicardipine. These channels have been demonstrated to be important for SMCs but not ICCs, so one might expect to see continued transients in the core ICCs but a loss of them in SMCs (Lee et al., 1999; PMID: 10444456)

We appreciate these comments. We have accordingly conducted new experiments with Nifedipine. Contrary to the expectation, Nifedipine ceases not only organoidal contractions, but also ICC activities (and its resulting synchronization) (new Fig. 7). These findings actually corroborate our model already mentioned in the first version that ICCs receive mechanical feedback from SMC's contraction to stably maintain their oscillatory rhythm. We believe that the additional findings with Nifedipine have improved the quality of our paper. Concerning the central cells in the organoid, we have additionally used anti-desmin antibody known to mark differentiated SMCs. Desmin signals perfectly overlap with those of aSMA in the peripheral single layer, supporting that the peripheral cells are SMCs and central cells are ICCs. The anti c-Kit antibody used in this study is what we raised in our hands by spending years (Yagasaki et al., 2021)), in which the antibody was carefully validated in intact guts of chicken embryos by multiple methods including Western Blot analyses, immunostaining, and in situ hybridization.

ANO1/TMEM16 are known to stain ICCs in mice. Antibodies against ANO1/TMEM16 available for avian specimens are awaited.

(9) Despite Tuj1+ enteric neurons only making up a small fraction of the organoids, the authors should still functionally test whether they regulate any aspect of contractility by treating organoids with an inhibitor such as tetrodotoxin to rule out a role for them.

Thank you for these advices, which are also raised by other reviewers. We have conducted TTX administration (new Fig. S2C). Changes in contractility by this treatment is not detected, supporting the argument that neural cells/activities are not essential for rhythmic contractions of the organoid (line 178-181).

(10) Finally, the manuscript is written to suggest that the focus of the study is to establish a system to interrogate ICC-SMC interactions in gut physiology and peristalsis. However, the organoids designed in this study are derived from the fetal precursors to the adult cell types. Thus, they might not accurately portray the adult cell physiology. I don't believe that this is a downfall, but rather a strength of the study that should be emphasized. That is, the focus could be shifted toward stressing the power of this new system as a reductionist, self-organizing model to examine the developmental emergence of contractile synchronization in the intestine - in particular that arising through ICC-SMC interactions.

We appreciate these advices. In the revised MS, we are careful so that our findings do not necessarily portray the physiological functions in adult gut.

Minor:

More technical information could be used in the methods:

(1) What concentration of Matrigel is used for coating, and what size were the wells that cells were deposited into?

We have added, "14-mm diameter glass-bottom dishes (Matsunami, D11130H)" and "undiluted Matrigel (Corning, 354248) at 38.5°C for 20 min" (line 471473).

(2) How were organoids transferred to the hydrogels? And were the hydrogels coated?

We have added “Organoids were transferred to the hydrogel using a glass capillary” (line 560-561).

(3) Tests for significance and *p* values should be added where appropriate (e.g. Figure S3B).

We have added these in Figure legend of new Fig. S3.

Reviewer #3 (Recommendations For The Authors):

This is an exciting study, and while the majority of our comments are minor suggestions to improve the clarity and impact of findings, it would be important to verify the effective disruption of GAP junction function with CBX or 18Beta-GA treatments before concluding they are not required for coordination of contractility and initiation by ICCs. It is possible that sufficient contextual support exists in the literature for the nature of treatments used, but this may need to be conveyed within the manuscript to allay concerns that the results could be explained by ineffective inhibition of GAP junctions.

Thank you very much for these advices. In the revised version, we have newly carried out experiments with dissociated embryonic heart cells cultured in vitro, a model widely used for gap junction studies (Fig. S3D). Both CBX or 18b-GA exert efficient inhibiting activity on contractions of heart cells. We have added the following sentence, “The inhibiting activity of the drugs used here was verified using embryonic heart culture (line 237-239)”.

<https://doi.org/10.7554/eLife.97860.2.sa0>